28 May 2026 19:34:47 ET | 46 pages

Hon Precision (7769.TW)

Initiate at Buy: Handling the Evolution in AI Chip Testing

Deep Dive

CITI'S TAKE

We view Hon Precision as a pure play capturing the multi-year testing upcycle. It is a key beneficiary of expanding global AI chips demand across AI GPU/ASIC, CPU, and networking ICs. Currently >70% of sales exposure is from AI/HPC segments, where it has secured the most leading AI projects, including Nvidia's AI GPU, Google TPU, AMD CPU, through its strong presence in final test (FT) handler market. Also, co-packaged optics (CPO) development should bring new TAM opportunities beyond '27E. We forecast strong overall sales/earnings CAGRs of 70%/77% (2025-28E). Our '27E/28E EPS forecasts are 6%/9% above consensus as we expect faster capacity ramp and content growth potential. Initiate at Buy; NT$11,000 target price.

Global No. 1 ATC handler provider — Active thermal control (ATC) handler picks & places then sorts chips in the FT process, which requires high thermal control, and Hon has a dominant >90% share in the AI/HPC FT market. Hon's unique position has allowed it to secure multiple AI projects with visibility into 2028. With the scaling of AI chips and extending testing time, we expect global OSATs to accelerate their testing capacity buildup, which should favor Hon's ATC handler shipment outlook.

More content value growth ahead — Besides a strong shipment pipeline, we also anticipate structural content growth driven by rising thermal requirement, with AI chips migration and growing AOI adoption by key GPU/ASIC customers, higher mix toward customized cold-palates products, and new tray specs adoption in coming years. We forecast Hon to achieve 10-15% ASP expansion per annum toward 2028E. And Hon's early development in CPO Insertion 4 O/E and MCL solution should allow it to unlock more long-term growth in evolving packaging architecture in the AI era.

50% CAGR capacity expansion — Given the promising visibility, we believe the binding constraint on Hon's outlook is capacity, not demand. Even with +40% capacity this year, it still can't fulfill its orders on hand. Thus Hon has been actively looking for new facility space, and set a 50% CAGR target through 2028 and beyond. We believe any new facility/land acquisition could support further share upside.

TP: NT$11,000 — Our TP is set at 36x PER on avg '27E/28E EPS. While shares have more than tripled in past 12 months, global peers are at 30-40x '27E/28E PER. We believe Hon's stronger c77% earnings CAGR (peers average c40%) and low QFII holding at 20% (peers at 34%) should still drive further share price re-rating. Key risks are slower-than-expected capacity addition, unexpected AI chip weakness.

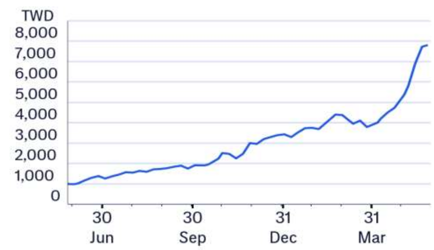
Earnings Summary

Year to 31 Dec	Net Profit (NT$M)	Diluted EPS (NT$)	EPS growth (%)	P/E (x)	P/B (x)	ROE (%)	Yield (%)
2024A	5,286	32.63	na	na	92.8	na	na
2025A	12,362	76.30	133.9	102.2	21.9	34.5	0.2
2026E	21,988	122.22	60.2	63.8	17.5	31.8	0.4
2027E	38,856	215.98	76.7	36.1	11.8	39.0	0.8
2028E	68,127	378.69	75.3	20.6	7.5	44.5	1.4

Source: Powered by dataCentral

Buy	
Catalyst Watch: Upside	
Price (28 May 26 13:30)	NT$7,700.00
Target price	NT$11,000.00
Expected share price return	42.9%
Expected dividend yield	0.4%
Expected total return	43.3%
Market Cap	NT$1,385,461M
	US$44,033M

Price Performance (RIC: 7769.TW, BB: 7769 TT)



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See Appendix A-1 for Analyst Certification, Important Disclosures and Research Analyst Affiliations.

7769.TW: Fiscal year end 31-Dec

Price: NT$7,800.00; TP: NT$11,000.00; Market Cap: NT$1,403,454m; Recomm: Buy

Profit & Loss (NT$m)	2024	2025	2026E	2027E	2028E
Sales revenue	13,992	30,271	51,615	88,350	149,816
Cost of sales	-6,293	-13,157	-21,866	-35,874	-58,308
Gross profit	7,700	17,114	29,749	52,476	91,509
Gross Margin (%)	55.0	56.5	57.6	59.4	61.1
EBITDA (Adj)	**6,330**	**15,107**	**26,715**	**48,016**	**84,927**
EBITDA Margin (Adj) (%)	45.2	49.9	51.8	54.3	56.7
Depreciation	-46	-62	-171	-293	-271
Amortisation	0	0	0	0	0
EBIT (Adj)	**6,285**	**15,046**	**26,545**	**47,723**	**84,656**
EBIT Margin (Adj) (%)	44.9	49.7	51.4	54.0	56.5
Net interest	160	306	545	545	545
Associates	0	0	0	0	0
Non-Op/Except/Other Adj	239	232	621	708	668
Pre-tax profit	**6,683**	**15,584**	**27,711**	**48,976**	**85,869**
Tax	-1,397	-3,222	-5,723	-10,120	-17,742
Extraord./Min.Int./Pref.div.	0	0	0	0	0
Reported net profit	**5,286**	**12,362**	**21,988**	**38,856**	**68,127**
Net Margin (%)	37.8	40.8	42.6	44.0	45.5
Core NPAT	5,286	12,362	21,988	38,856	68,127
Per share data	**2024**	**2025**	**2026E**	**2027E**	**2028E**
Reported EPS ($)	32.63	76.30	122.22	215.98	378.69
Core EPS ($)	32.63	76.30	122.22	215.98	378.69
DPS ($)	0	14.37	33.61	59.78	105.63
CFPS ($)	14.24	100.95	99.91	173.26	310.17
FCFPS ($)	13.43	99.19	95.61	168.75	305.43
BVPS ($)	84.10	355.39	445.14	661.13	1,039.81
Wtd avg ord shares (m)	161	162	180	180	180
Wtd avg diluted shares (m)	162	162	180	180	180
Growth rates	**2024**	**2025**	**2026E**	**2027E**	**2028E**
Sales revenue (%)	na	116.3	70.5	71.2	69.6
EBIT (Adj) (%)	na	139.4	76.4	79.8	77.4
Core NPAT (%)	na	133.9	77.9	76.7	75.3
Core EPS (%)	na	133.9	60.2	76.7	75.3
Balance Sheet (NT$m)	**2024**	**2025**	**2026E**	**2027E**	**2028E**
Cash & cash equiv.	6,025	53,427	71,482	101,809	156,725
Accounts receivables	2,913	2,935	5,141	9,426	16,070
Inventory	8,444	12,641	18,866	32,077	51,264
Net fixed & other tangibles	2,234	2,810	2,608	3,127	3,709
Goodwill & intangibles	29	91	119	119	119
Financial & other assets	2,221	1,366	4,408	10,240	19,283
Total assets	**21,866**	**73,270**	**102,624**	**156,798**	**247,170**
Accounts payable	3,181	3,402	5,476	9,878	16,271
Short-term debt	157	0	0	0	0
Long-term debt	0	0	0	0	0
Provisions & other liab	4,796	11,837	17,066	27,981	43,834
Total liabilities	**8,134**	**15,239**	**22,542**	**37,859**	**60,105**
Shareholders' equity	13,732	58,031	80,082	118,939	187,066
Minority interests	0	0	0	0	0
Total equity	**13,732**	**58,031**	**80,082**	**118,939**	**187,066**
Net debt (Adj)	**-5,868**	**-53,427**	**-71,482**	**-101,809**	**-156,725**
Net debt to equity (Adj) (%)	-42.7	-92.1	-89.3	-85.6	-83.8

Valuation ratios	2024	2025	2026E	2027E	2028E
PE (x)	na	na	63.8	36.1	20.6
PB (x)	92.8	21.9	17.5	11.8	7.5
EV/EBITDA (x)	na	90.9	50.2	27.4	15.0
FCF yield (%)	0.2	1.3	1.2	2.2	3.9
Dividend yield (%)	na	0.2	0.4	0.8	1.4
Payout ratio (%)	0	19	27	28	28
ROE (%)	na	34.5	31.8	39.0	44.5
Cashflow (NT$m)	**2024**	**2025**	**2026E**	**2027E**	**2028E**
EBITDA	6,330	15,107	26,715	48,016	84,927
Working capital	-5,641	3,900	-4,217	-8,010	-12,629
Other	1,618	-2,652	-4,525	-8,835	-16,497
Operating cashflow	**2,308**	**16,356**	**17,973**	**31,171**	**55,801**
Capex	-133	-286	-773	-812	-852
Net acq/disposals	-413	-465	666	0	0
Other	-777	0	0	0	0
Investing cashflow	**-1,322**	**-751**	**-107**	**-812**	**-852**
Dividends paid	0	-2,586	-6,047	-10,756	-19,006
Financing cashflow	**-3,759**	**31,886**	**148**	**0**	**0**
Net change in cash	**-2,769**	**47,491**	**18,014**	**30,359**	**54,948**
Free cashflow to s/holders	**2,175**	**16,070**	**17,200**	**30,359**	**54,948**

For definitions of the items in this table, please click here.

Contents

Investment Strategy

Initiate at Buy

We initiate on Hon Precision with a Buy rating as we view Hon Precision sitting in a structurally protected position at the heart of the AI/HPC back-end test cycle, with growth visibility extending well into 2028E — The company holds >90% share of the AI/HPC final-test handler market and 60-70% of the ASIC SLT market. We forecast a solid c70% sales revenue CAGR through 2025-28E with gross margin expanding to the >60% range, and in our view the binding constraint on Hon Precision's near-term trajectory is now capacity, rather than demand — placing the company in the early phase of a multi-year structural growth and margin-expansion cycle.

Our NT$11,000 target price is set at a target 36x PER on 2027E/28E average EPS. While the shares have more than tripled over the past 12 months, compared to global peers trading at 30-40x 2027E/28E PER, we believe additional upside is very feasible, with Hon's stronger c77% earnings CAGR (vs. peers' average c40% CAGR) and relatively low QFII holding at 20% (vs. peers at 34%) still driving further share price re-rating.

Competitive edge and strategy

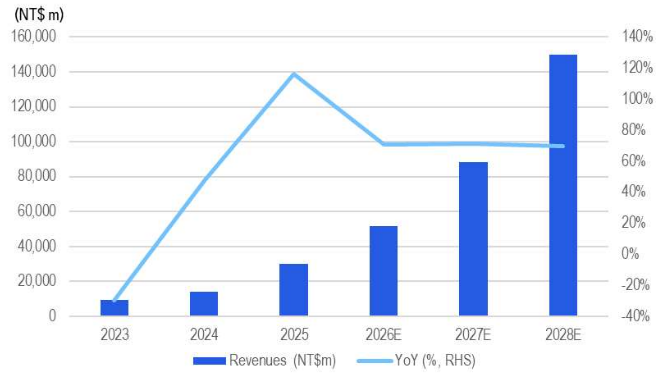
The company's current business mix tilts decisively toward the AI/HPC end of the market: AI GPU, ASIC, networking-switch and related HPC chipsets together accounted for 78% of 1Q26 order intake, with automotive and mobile AP at roughly 9% each. Early-2010s breakthroughs at Amkor and Qualcomm, and later openings into Apple, Broadcom, MediaTek and major US cloud-service providers, set the stage for that mix — but the present-day commercial profile is fundamentally an AI-test profile.

Backed by a formidable global intellectual property estate exceeding 600 granted patents as of 2025, Hon Precision's engineering capability spans multi-zone cold plate architectures, advanced thermal control algorithms, high-precision socket engagement pressure control, and specialized condensation mitigation. This deep technological foundation underpins an aggressive innovation pipeline. R&D investments increased by 63% year-over-year in 2025, and management is accelerating this momentum through 2026 to spearhead breakthroughs in large-package next-generation platforms, 3D automated optical inspection (AOI) enhancements, micro-channel liquid (MCL) handler development, and CPO Insertion 4 initiatives.

This technical investment directly drives Hon Precision's product strategy. Structurally, it prioritizes modular systems engineered for rapid maintenance—a critical operational advantage for OSATs looking to mitigate costly downtime during high-volume manufacturing runs. Furthermore, to maximize market versatility, the company's handlers and ATC solutions feature universal compatibility with all major ATE testor platforms, deliberately avoiding the constraints of a single-tester ecosystem. Ultimately, Hon Precision secures its premium market positioning and strong pricing leverage by isolating its focus on the industry's most formidable mechanical and thermal test automation bottlenecks. By tackling the complex high-end challenges that intensify with every new chip node, the company successfully insulates itself from the margin-eroding commodity pressures faced by lower-tier handler manufacturers.

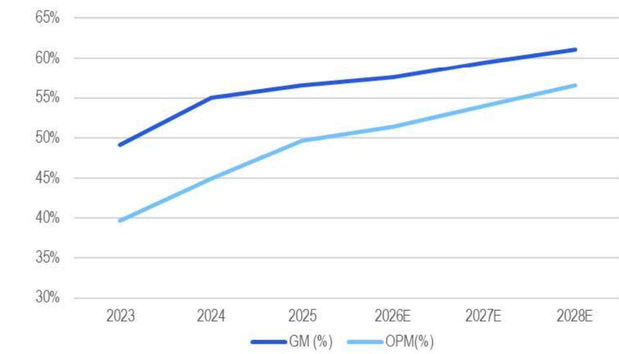
Key charts

Figure 1. Hon – Revenues (NT$ mn) & YoY growth



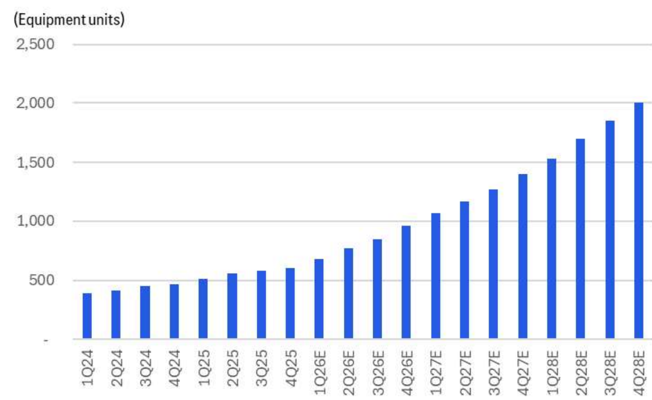
Source: Citi Research, Company Reports

Figure 2. Hon – GM and OPM trend



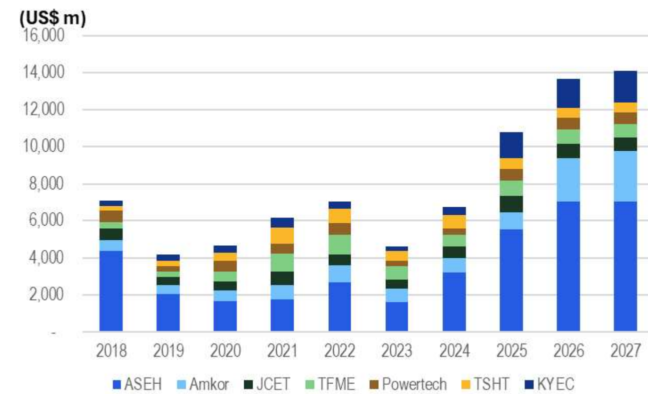
Source: Citi Research, Company Reports

Figure 3. Hon – capacity expansion forecast



Source: Citi Research, Company Reports

Figure 4. Key global OSATs capex trend



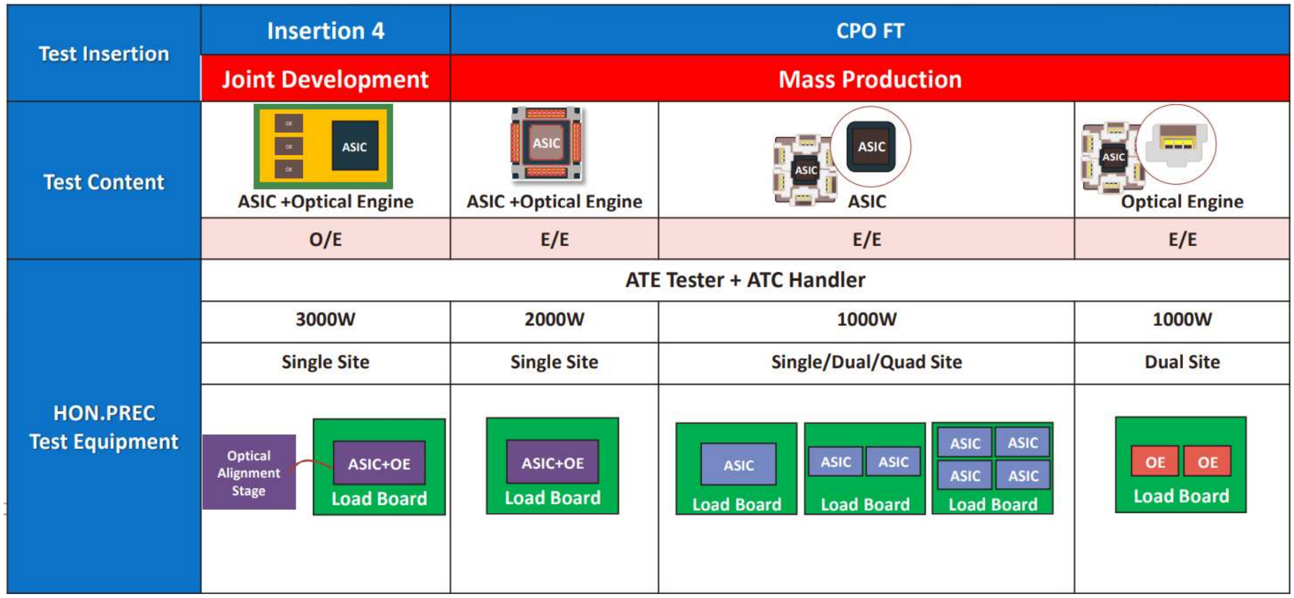
Source: Citi Research, Company Reports

Figure 5. Key testing vendors for AI GPU/ASIC, CPU

Customer	ASIC vendor	Application	Chips	Production time	CP	Probe card vendor	FT/Burn-in/SLT	FT Testor	FT Handler	FT Socket	SLT Handler	SLT Socket
Nvidia		AI GPU	Rubin	2H26	TSMC	Technoprobe	KYEC/ASEH	Advantest	Hon	Winway	Chroma	Smith/Winway
		AI CPU	Vera	2H26	ASEH	WinWay/MPI/Technoprobe	KYEC	Advantest	Hon	Winway	Chroma	Smith/Winway
		AI Switch	Spectrum/Quantum	2H26	ASEH	MPI	KYEC	Advantest/Teradyne	Hon	Winway	Chroma	Smith/Winway
AMD		AI GPU	MI400 series	2H26	TSMC	Techoprobe	ASEH	Advantest	Hon	Winway	Hon/Chroma	Smith/Winway
		AI CPU	Venice	2H26	TSMC/ASEH	Techoprobe/MPI	ASEH	Advantest	Hon	Winway	Hon/Chroma	Smith/Winway
Google	Broadcom	AI ASIC	TPU v8i	2H26	TSMC/KYEC/Ardentec	MPI	KYEC	Advantest	Hon	LEENO	na	Winway
	Mediatek	AI ASIC	TPU v8t	2H26	TSMC/KYEC	Technoprobe	KYEC	Advantest	Hon	Winway	Hon	Winway
	Mediatek	AI ASIC	TPUv9	2H27	TSMC/KYEC	Technoprobe/KSMT/MPI	KYEC	Advantest	Hon	Winway	Hon	Winway
	GUC	AI CPU	Axion	1H26	TSMC/KYEC	MPI		Advantest	Hon	WinWay	Hon	Winway
AWS	Marvell	AI ASIC	Trianium 2	2H24	TSMC/Ardentec	MPI	ASEH	Advantest	Hon	LEENO	na	LEENO
	Alchip	AI ASIC	Trianium 3	2H26	TSMC	MPI	ASEH/Terapower	Advantest/Teradyne	Hon	LEENO	na	LEENO
Tesla	GUC	AI ASIC	AI5	2H27			Doosan/KYEC	Advantest/Teradyne	Hon	Winway	Hon	Winway

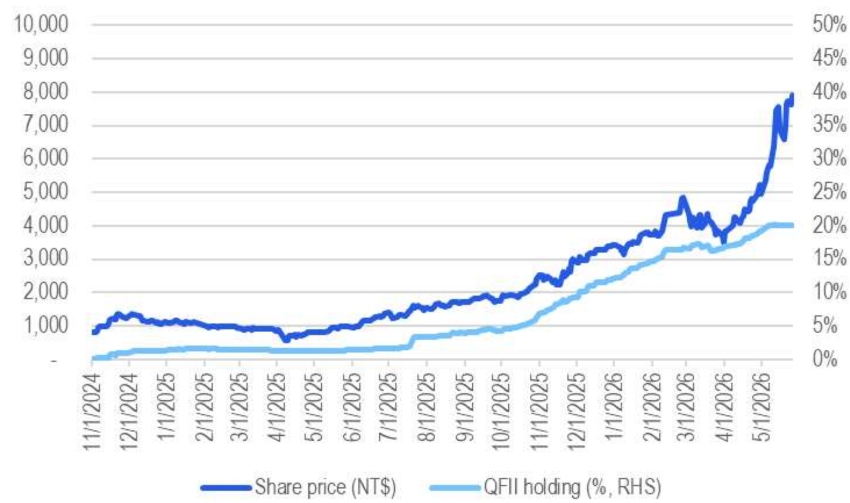
Source: Citi Research

Figure 6. Hon - CPO testing (Insertion 4 and CPO FT)



Source: Citi Research, Hon Precision

Figure 7. Hon – further share price re-rating potential as QFII holding is still low at 20%



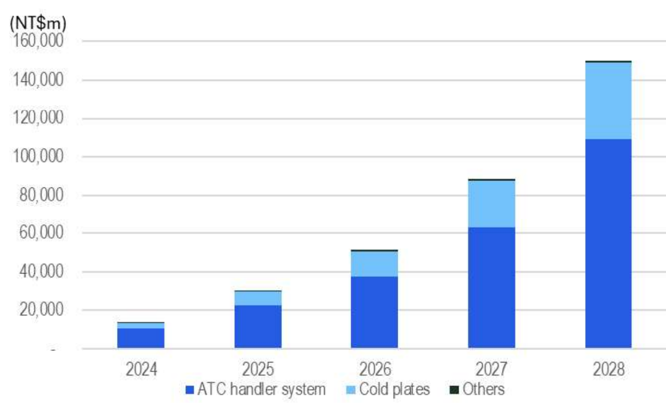
Source: Citi Research, TEJ

Global Leading Position

Global leading position in AI/HPC back-end testing equipment

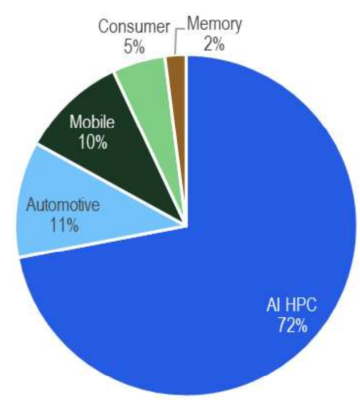
Hon Precision primarily provides testing equipment bundled with highly customized thermal control systems for global OSAT customers. Its main products, ATC handlers, are used in semiconductor final-test (FT) and system-level test (SLT) processes. In 2025, around 75% of its revenue came from ATC handler system and a low-20% came from its cold plates product line. The company usually provides them as a bundled package rather than as discrete parts, ie, a handler is paired with its matching ATC system and cold plate/SLK that interfaces with the chip.

Figure 8. Hon – revenue breakdown by product



Source: Citi Research, Hon Precision

Figure 9. Hon – revenue breakdown by application (2025)



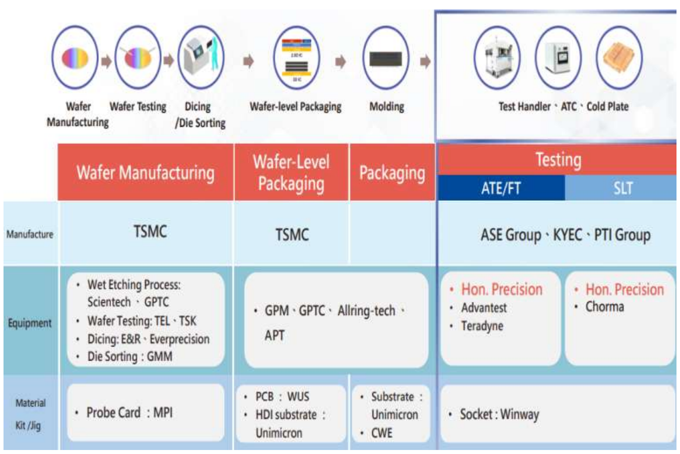
Source: Citi Research, Hon Precision

For its pairing of **handler+ATC**, what goes into that bundle depends on how thermally demanding the end-device is. There are mainly three type of bundle configurations:

- **Tri-temp ATC+handler** (accounting for 20% of Hon's total ATC handler shipment; ASP at NT$15-18m): this is mainly adopted by automotive and aerospace applications, which require the most demanding thermal control systems
- **Dual-temp ATC+handler** (75% shipment; ASP at NT$12-15m): mainly adopted by high-computing applications, including AI GPUs, ASICs and server CPUs
- **Handler only** (5% shipment; ASP at around NT$6m): used more by consumer-related applications

Working concept of ATC handler: In the chips FT testing process, a test head (extending from the ATE testor, provided by Advantest or Teradyne) carries the load board and the test socket, translating ATE's electronics signals and contacting with device under test (DUT) to implement testing. The job of the handler (mostly provided by Hon for AI/HPC chips) is mainly for the transportation of the DUT, and the ATC serves as an active thermal control system allowing the DUT to be tested under the required temperature. Especially for AI/HPC handlers, the importance of ATC+handler is structurally increasing as it is responsible for four functions in parallel. First, it must precisely **transport** packaged ICs from the input tray to the test socket with high requirement of effective throughput. Second, it must dynamically control each device under the required test **temperature** ahead of and during measurement. Third, it must deliver precise **mechanical contact force** when pressing the IC onto the test socket, which gets more critical as AI chips carry thousands of contact pins at sub-millimetre pitch. Fourth, once the tester returns a result, it **sorts** each device to the appropriate output category according to its performance grade. Note that these four functions are not modular — they must operate as a tightly integrated system, with thermal control, mechanical handling, and electrical signal delivery synchronized within the test pattern timing of the ATE testor.

Figure 10. Semiconductor front-end/back-end process flow



Source: Citi Research, Hon Precision

Figure 11. Demo of ATE testor working with handler

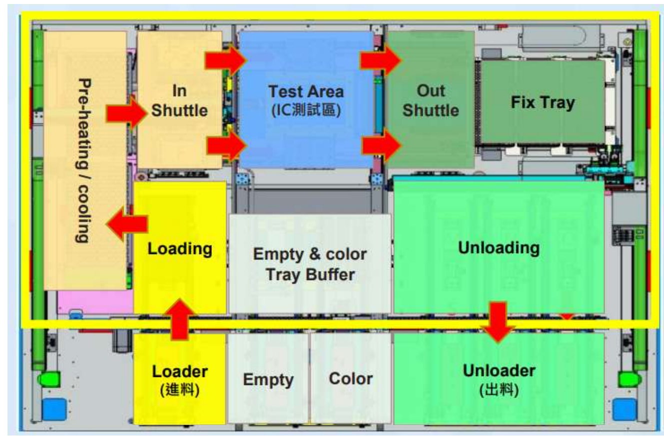
Source: Citi Research, IEEE

What distinguishes Hon Precision in this space is its ability to hold chips at temperature when several kilowatts of heat are dissipating, which has become more critical in AI/HPC application as more dynamic thermal control is required. Hon's handlers are also responsible for keeping contact force stable as packages get larger and heavier, and delivering mechanical placement accuracy tight enough that yield and repeatability are not the limiting factors at the test cell. Moreover, Hon Precision does not compete in the tester market itself, which is dominated by Advantest and Teradyne, whose flagship platforms (such as V93000) carry more than 3x the dollar content of a handler. By design, Hon Precision's handlers are compatible with all major ATE testors, allowing the company to enter into any tester ecosystem without forcing customers to choose.

Figure 12. Hon's handler paired with ATC system

Source: Citi Research, Hon Precision

Figure 13. Hon's handler tray work flow



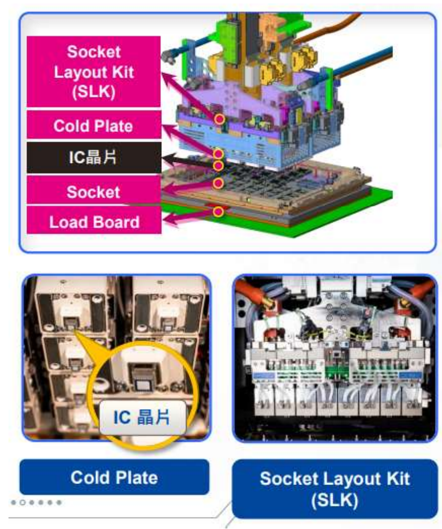
Source: Citi Research, Hon Precision

Figure 14. Hon Precision – overview of key product offerings

Source: Citi Research, Hon Precision

For its **cold plates** products, they currently contribute low-20% of Hon's total revenue. They serve as a recurring, margin-accretive consumable stream that scales with the installed base. It is a highly engineered thermal interface that sits at the bottom of the ATC thermal head, coming into direct, high-pressure contact with the packaged device under test (DUT). Cold plates replacement demand is driven by customer product generation, since each chip requires a differently shaped cold plate. At initial equipment installment, Hon Precision bundles cold plates with handler shipments (around two cold plates for large AI chips). This is why cold-plate revenue scales structurally with the handler installed base. A handler is a long-duration capital asset that stays in service for the better part of a decade, whereas the cold plate is a chip-specific consumable that turns over far more frequently.

Figure 15. Hon's cold plate products

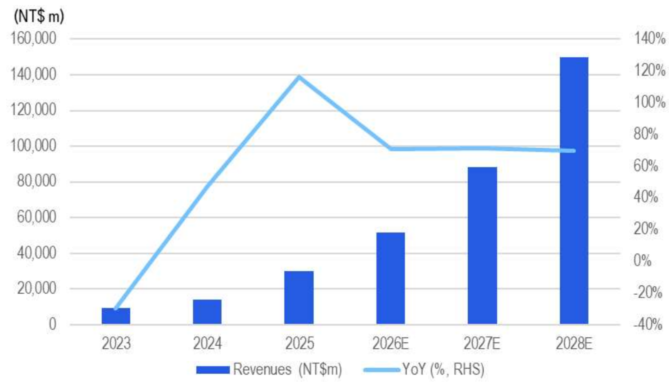


Source: Citi Research, Hon Precision

Major beneficiary of expanding AI/HPC testing market

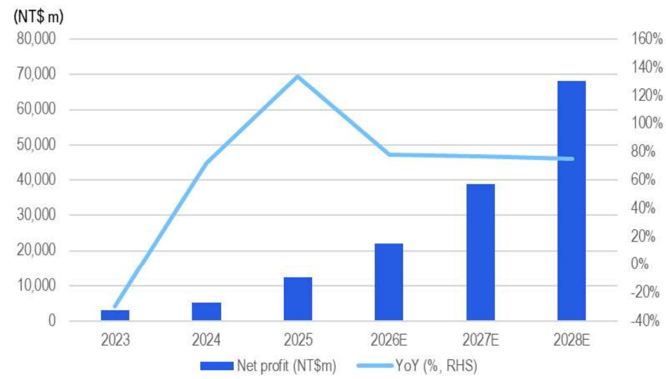
We view Hon Precision's high market share in AI/HPC projects positioning it as a pure play to capture the multi-year buildup for AI chips testing. In the final test (FT) ATC handler market, Hon Precision has a dominant share (>90%) of global AI/HPC projects. In system-level test (SLT), it has around 30% share among AI/HPC projects, with Chroma leading the SLT market for AI/HPC projects. Given Hon's high share in major AI projects and an expanding TAM from the emergence of AI ASIC, CPU and networking, we forecast Hon's revenue to increase nearly five-folds from 2025 to 2028, or a strong CAGR of ~70%. With an increasing mix of high-end ATC handler and cold plates sales, we also expect solid margins expansion for Hon over the same period, driving an estimated earnings CAGR of 77%.

Figure 16. Hon – total revenue (NT$ mn) & YoY growth



Source: Citi Research, Company Reports

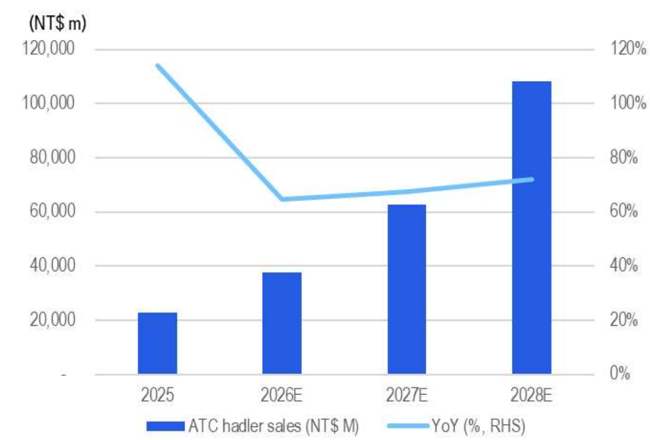
Figure 17. Hon – total net profit (NT$ mn) & YoY growth



Source: Citi Research, Company Reports

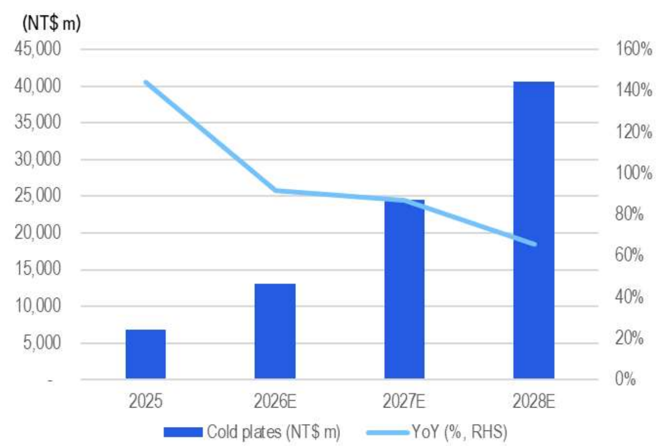
By product line, we forecast **ATC handler** sales to grow at a >65% CAGR through 2025-28, with ASP expansion of 10-15% per annum and shipment growth of a ~50% CAGR. For **cold plates**, due to stronger replacement demand in a rapid chips generation cycle and ASP expansion on more customized designs, we forecast cold plats sales to register around an 80% sales CAGR towards 2028.

Figure 18. Hon – ATC handler revenue (NT$ mn) & YoY growth



Source: Citi Research, Company Reports

Figure 19. Hon – cold plates revenue (NT$ mn) & YoY growth



Source: Citi Research, Company Reports

The structural growth baseline for Hon Precision is the continued scaling of AI chips, which drives a faster buildup of industry back-end testing capacity. Given expanding chips demand from AI GPU to AI ASIC, CPU and networking, we believe back-end testing has become one of the supply chain constraints. Thus, global major OSATs, including ASEH, Amkor and KYEC, etc. have all announced aggressive capacity expansion plans in recent years. And we note that most of their capex planning for this year is for clean room/facility purchases, which we view as a leading indicator of stronger equipment pull-in in the upcoming years.

Specification migration driving structural ASP and content uplift. As AI chip thermal design power (TDP) has continued to climb, Hon Precision's customers have upgraded along the ATC ladder. The result is that Hon Precision is capturing more dollars per handler with each successive AI chip generation, on top of the unit-volume growth — a dynamic that has contributed materially to both revenue growth and gross margin. Moreover, the evolving packaging architecture, including larger packaging size, CPO adoption and add-on functions, including the AOI integration, should create an substantial content value uplift for Hon's long-term growth potential, in our view.

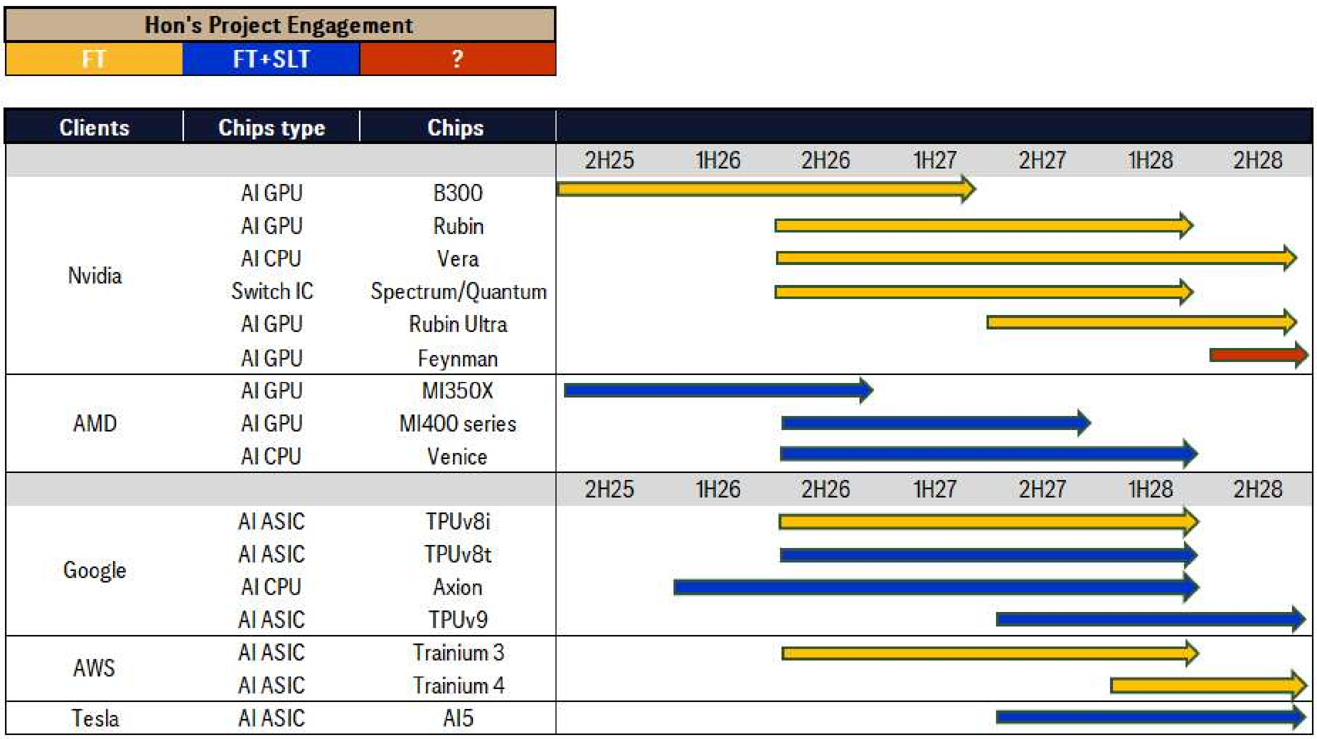
Promising AI/HPC pipeline underpins +50% CAGR capacity expansion

An aggressive LT capacity expansion scope

Hon's capacity guidance has been raised twice within two quarters, signaling demand visibility through year-end — Hon Precision's 2026 capacity expansion target was originally set at +30% YoY guided in 4Q25, then was raised to +40% YoY in 1Q26 in response to stronger-than-expected order intake. At the earnings call in May 2026, management indicated that even the revised +40% level cannot fully satisfy current customer demand, and that actual expansion could exceed +40% following the recent acquisition of a new plant. The 2026 expansion is supported by a 50% YoY increase in total workforce, and the acquisition of its new Daya plant (production likely to come online in 4Q26 with around +25% new plant space addition, compared to current capacity) — a combination we read as a high-conviction signal of demand visibility. Looking further, management disclosed an internal project to expand 2027 capacity by an additional ~50% vs. 2026 to meet customer demand, with the company still actively searching for additional large plant space nearby.

Given Hon's tight relationship with clients and industry-leading equipment performance, it has secured multiple AI chips projects with global major customers. Promising project visibility and order booking have reinforced Hon's ability to continue its strong expansion, and we believe the major constraint for Hon's near-term growth is whether it can secure more new production space and workforce. Regarding equipment delivery schedule, typically, a new round of testing/handler equipment buildup plan starts shipment 1-2 quarters ahead of chips volume production.

Figure 20. Chips shipment schedule for Hon's major AI customers



Source: Citi Research

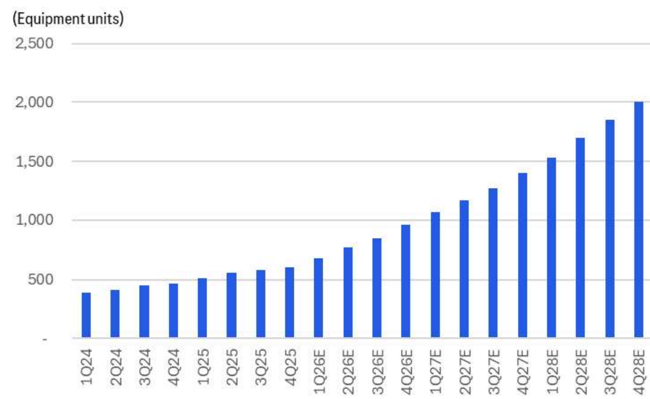
Hon Precision currently operates three core production plants in Taichung plus a Suzhou (China) facility, supplemented by additional leased space. Before its Longshan plant starts to contribute (only scheduled in 2028), Hon Precision is bridging the supply/demand gap leveraging its recent acquisition of another Taichung Daya plant, with production likely to come online in 4Q26. This newly acquired plant was previously a machine-tool factory, shares part of the manufacturing process, and offers 20-30% more floor space than an existing plant, with layout modifications required before ramp. As Hon is still actively searching for new land space or lease plants for further capacity addition, we believe any new announcement of new facility acquisition could drive upside surprise to Hon's near-term growth outlook.

Figure 21. Hon's key production plants

Plants	Facility space	MP time	Note
Taichung Plant 1	11,400m2	In production	
Taichung Plant 2	12,400m2	In production	
Taichung Plant 3	9,500m2	In production	
Suzhou Plant	3,000m2	In production	
Taichung Plant 4 (Longshan)	18,000m2	2028	Under construction
Taichung Plant 5 (Newly accquired)	~10,000m2	4Q26-1Q27	Only layout modifications required

Source: Citi Research, Hon Precision

Figure 22. Hon – capacity expansion forecast



Source: Citi Research, Hon Precision

Added AOI function to lift content value in Hon's handlers

Hon Precision recently introduced the Top/Bottom Automated Optical Inspection (AOI) functionality — which received customer qualification in early 2026 — representing an emerging ASP and volume driver. AOI integration combines mechanical handler functions with in-line optical defect inspection, allowing the test cell to detect surface defects, marking errors and structural anomalies on packaged devices without requiring a separate inspection station. Per management, AOI integration lifts FT handler ASP by 10-15% per system, with the current order book standing at about 800 units and the 2026 installed base expected to exceed 1,000 sets. Management has formally indicated AOI integration as one of the core 2026-27 growth drivers.

We see four structural drivers behind AOI's emergence as a must-have feature for high-end FT handlers. First, AI/HPC packages are becoming so expensive that any post-test defect detection that catches a flawed device earlier in the workflow has strong yield-economic justification. Second, the rise of advanced packaging means visible defects, such as cracks, mark misalignment and lid defects, are more consequential than they were for monolithic devices. Third, hyperscaler customers are increasingly demanding full-automation and integrated architectures where AOI inspection is integrated directly into the test handler.

In our base case scenario, we assume an adoption rate of around 30% of Hon's handler system, mostly adopted by high-end HPC, in 2026. With higher integration requirement, throughput efficiency, more advanced packaging solution in next-generation AI chips, we expect the adoption rate to gradually increase in the coming years. While order shipment may occur this year, meaningful revenue contribution is more likely in 2027, in our view, given the 2-3 quarter lag in revenue recognition after equipment shipment common in the industry. We thus estimate the add-on AOI feature could bring an incremental ASP uplift of c5/3% in 2027/2028. Overall, we view AOI integration providing immediate ASP uplift and margin accretion while the large-package and CPO Insertion 4 features should drive long-term strong growth opportunities.

Figure 23. Hon – newly released FT handler

Source: Citi Research, Hon Precision

Strong competitive edges

Industry-leading thermal management capability, ecosystem lock-in, and testor compatibility

First, the company has the industry's broadest dual-temperature and tri-temperature **ATC (active thermal control)** product portfolio. ATC has been a Hon Precision focus since the early 5G infrastructure cycle — the company committed engineering resources to the technology before most of the industry recognized it as a critical specification. The current commercial line-up handles devices up to 6kW thermal load (ATC3.7), with the ATC3.8 (10kW dual-temp) and ATC5.7 (10kW tri-temp) platforms scheduled for 2027-28 introduction.

Second is its **universal tester compatibility**. Hon Precision's handlers are designed to be attached to all major ATE testors — Advantest V93000, Teradyne UltraFLEX, and the various custom platforms used by hyperscaler customers — without forcing the customer into a single tester ecosystem. By contrast, regional handler peers in general focus on narrower tester ecosystems, which restrict their addressable demand. Hon benefits from total AI chip test volume regardless of which OSAT, which tester vendor, or which chip designer wins individual programs.

The last edge, in our view, is its **installed-base lock-in at major OSATs**. Hon Precision's installed base among Taiwan-based OSATs (KYEC, ASEH, etc.) is a multiple of any competitor's cumulative handler installed base. That installed base compounds the company's advantage through three factors: the density of field application engineers stationed at customer sites, the depth of spare-parts inventory and the accumulated integration know-how that allows new ATC handler models to be qualified quickly against existing test programs. Typically, customers in advanced FT segments incur a lengthy qualification process when adopting a new handler supplier — a friction that effectively helps lock in incumbents' positions for the duration of a chip generation. We believe replicating Hon Precision's entire OSAT-embedded service and installed-base ecosystem would take a competitor several years even with a technically competitive product.

Figure 24. Key testing vendors for AI GPU/ASIC, CPU

Customer	ASIC vendor	Application	Chips	Production time	CP	Probe card vendor	FT/Burn-in/SLT	FT Testor	FT Handler	FT Socket	SLT Handler	SLT Socket
Nvidia		AI GPU	Rubin	2H26	TSMC	Technoprobe	KYEC/ASEH	Advantest	**Hon**	Winway	Chroma	Smith/Winway
		AI CPU	Vera	2H26	ASEH	WinWay/MPI/Technoprobe	KYEC	Advantest	**Hon**	Winway	Chroma	Smith/Winway
		AI Switch	Spectrum/Quantum	2H26	ASEH	MPI	KYEC	Advantest/Teradyne	**Hon**	Winway	Chroma	Smith/Winway
AMD		AI GPU	MI400 series	2H26	TSMC	Techoprobe	ASEH	Advantest	**Hon**	Winway	**Hon/Chroma**	Smith/Winway
		AI CPU	Venice	2H26	TSMC/ASEH	Techoprobe/MPI	ASEH	Advantest	**Hon**	Winway	**Hon/Chroma**	Smith/Winway
Google	Broadcom	AI ASIC	TPU v8i	2H26	TSMC/KYEC/Ardentec	MPI	KYEC	Advantest	**Hon**	LEENO	na	Winway
	Mediatek	AI ASIC	TPU v8t	2H26	TSMC/KYEC	Technoprobe	KYEC	Advantest	**Hon**	Winway	**Hon**	Winway
	Mediatek	AI ASIC	TPUv9	2H27	TSMC/KYEC	Technoprobe/KSMT/MPI	KYEC	Advantest	**Hon**	Winway	**Hon**	Winway
	GUC	AI CPU	Axion	1H26	TSMC/KYEC	MPI		Advantest	**Hon**	WinWay	**Hon**	Winway
AWS	Marvell	AI ASIC	Trianium 2	2H24	TSMC/Ardentec	MPI	ASEH	Advantest	**Hon**	LEENO	na	LEENO
	Alchip	AI ASIC	Trianium 3	2H26	TSMC	MPI	ASEH/Terapower	Advantest/Teradyne	**Hon**	LEENO	na	LEENO
Tesla	GUC	AI ASIC	AI5	2H27			Doosan/KYEC	Advantest/Teradyne	**Hon**	Winway	**Hon**	Winway

Source: Citi Research

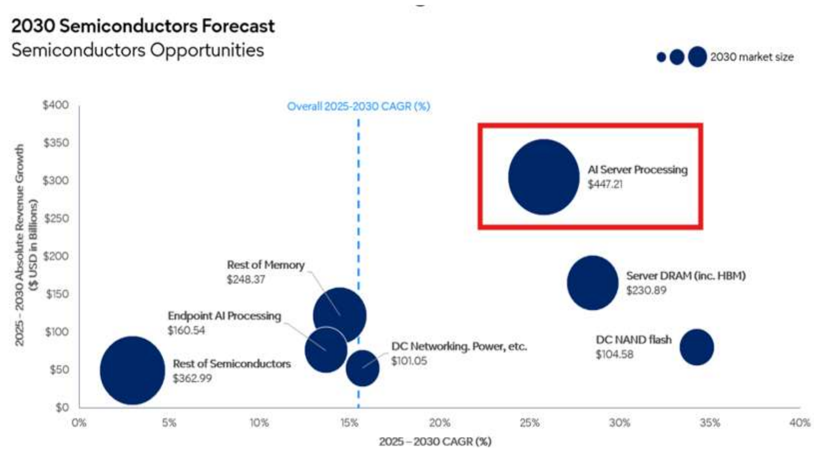
Secular Industry Trend 1

Handler Shipment Growth Driven by Expanding AI Chips Demand, Lengthening Testing Time

Expanding AI chips demand = faster back-end testing capacity buildup

The structural growth baseline for Hon Precision is the continued scaling of AI GPU, AI ASIC and data-center CPU shipments. We expect the expansion in AI chips shipments to be sustainable, underpinned by solid growth from the AI GPU camp with Nvidia having already stated that product demand visibility now extends through 2027 with projected DC sales of US$1T in 2025-27. Moreover, the demand base is broadening: whereas 2024-25 AI compute demand was concentrated in AI GPU, the 2026-27 demand mix increasingly includes data-centre CPU (NVIDIA Vera, AMD Venice, Google Axion) and a widening cohort of AI ASICs (Google TPU, AWS Trainium 3, Tesla AI5, etc.). Of note, given the increasing demand for agentive AI, our global semiconductor team recently published a CPU TAM model report, and expects the CPU TAM could grow strongly at a 35% CAGR through 2030. Within the expanding CPU TAM, we see agentic CPUs to grow at a strong 185% CAGR to US$59.4bn in 2030. Last but not least, the chip roadmap pipeline is dense and well-defined through 2028, providing multi-year visibility on test-equipment demand. This multi-year growth should bode well for Hon Precision as each incremental device category — regardless of designer — still requires an FT handler at the back end.

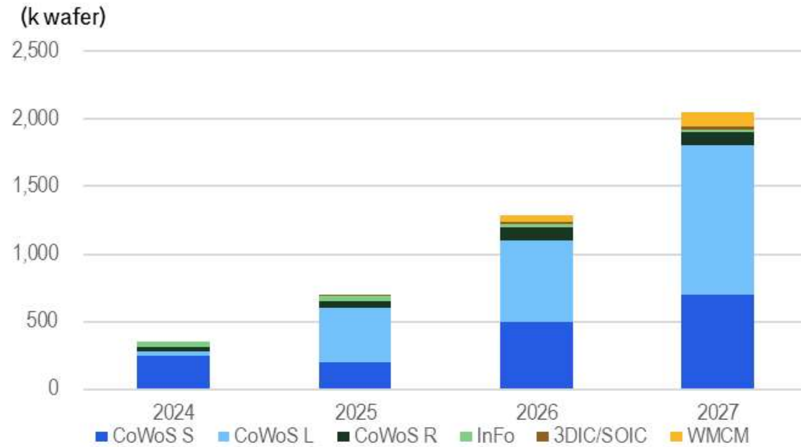
Figure 25. Global semiconductor Opportunities Through 2030



Source: Citi Research, Gartner

We believe one of the most reliable single proxies for AI chip volume is TSMC's CoWoS capacity, as virtually every leading-edge AI accelerator must transit CoWoS advanced packaging. We estimate CoWoS capacity could continue growing at nearly 85% YoY and 60% YoY in 2026 and 2027, respectively — and the growth could be further accelerated by the addition of OSATs' new CoWoS-like capacity buildout.

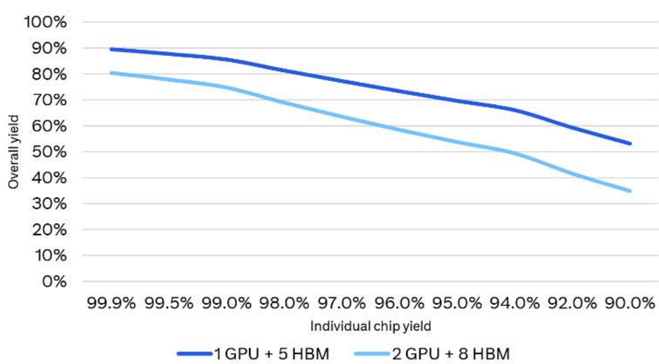
Figure 26. TSMC advanced packaging capacity forecast



Source: Citi Research estimates

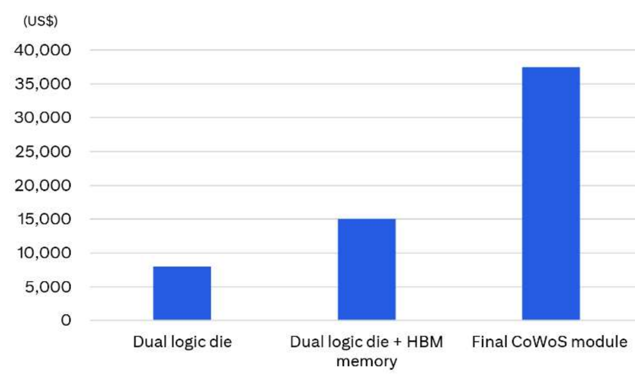
Semiconductor testing is becoming increasing critical in the AI era. In the monolithic-die era, semiconductor test was largely a back-end parametric check on an essentially finished product, and the manufacturing flow was tolerant of moderate test escape rates because the per-device economic stake was limited. Heterogeneous integration has fundamentally broken this model. A leading-edge AI accelerator now stacks 2 logic dies and 8 HBM stacks onto a single CoWoS-L interposer at a reticle size exceeding 3x, and the failure of any single constituent — a die, a micro-bump, an HBM interface, an interposer routing trace — causes the entire multi-thousand-dollar package to be scrapped. The economic value at risk per test decision has therefore risen. We see testing as having moved structurally from a downstream cost line to a primary determinant of whether AI infrastructure capacity can be delivered to hyperscale customers at volumes — a shift that explains why OSAT capex on test capacity is now growing in step with, rather than lagging, advanced packaging capacity.

Figure 27. Individual chip yield decline can lead to a sharp drop in overall chip yield



Source: Citi Research

Figure 28. Scrapped cost rises sharply through stages in advanced packaging



Source: Citi Research

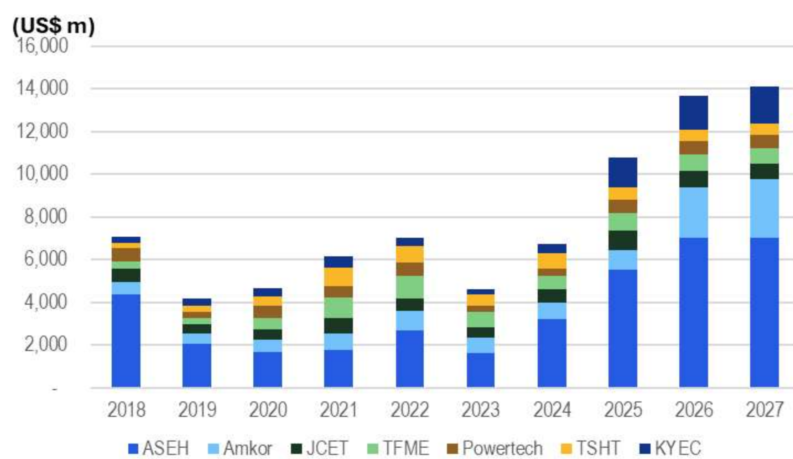
Rapid AI product cadence driving higher equipment buildout by OSATs

This enhances production throughput

The surging AI chip volume is also pushing OSATs to respond with record-high capex planning with major focus on leading-edge advanced packaging and testing. The combined 2026 capex of Taiwan's three major OSATs — ASEH, Powertech, and KYEC — is expected to reach more than NT$300bn, marking a third consecutive year of record-high OSAT investment. Capex allocation towards testing is rising at the same time. Also, we note that a significant part of most OSATs' capex budget this year is for land purchase/facility construction, implying that more heavy equipment pull-in would likely happen in the upcoming years. Following KYEC's AI GPU-related equipment build of nearly 700 units in 2025, it has also stated plans to install additional c1,000 units of equipment in 2026 and more in 2027 for AI GPU/ASIC demand. With more facility construction likely to be completed next year, we expect stronger equipment pull-in demand in the upcoming years as demand is broadening, ie, not only from AI GPU but also AI ASIC and CPU in the coming years.

The accelerating chip generation cycle is the second, often-overlooked, driver — each new generation forces OSATs to re-tool, rather than simply reuse existing capacity. Rising chip volume alone would justify a steady capacity expansion, but the AI generation cycle has compressed in a way that fundamentally changes the nature of the OSAT equipment build. The AI accelerator refresh cadence has shortened from the historical 2-3 year cycle to approximately 1-1.5 years — NVIDIA's roadmap alone runs Blackwell (2024), Blackwell Ultra (2025), Rubin (2026), Rubin Ultra (2027), and Feynman (2028) in consecutive annual or near-annual steps. Each generation is not merely a higher-volume version of its predecessor; it carries materially different package dimensions, thermal envelopes, and test requirements. The compressed generation cycle therefore multiplies the equipment build requirement: OSATs need to simultaneously expand capacity for the current generation and re-tool for the next.

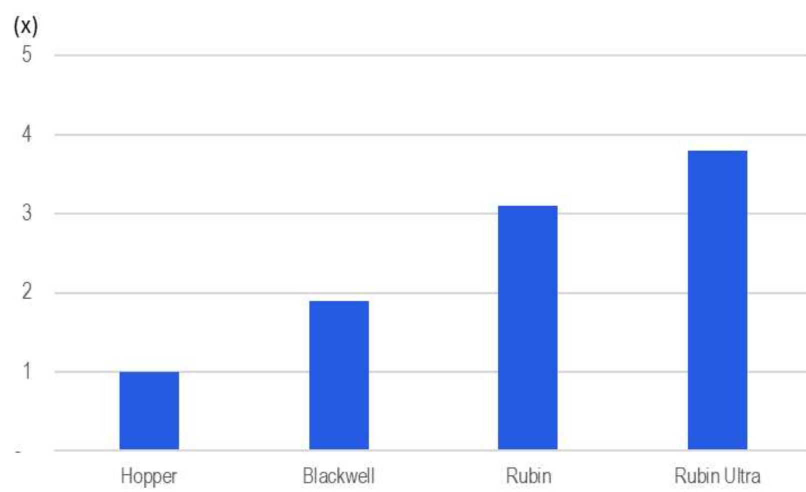
Figure 29. Global major OSAT capex



Source: Citi Research, Bloomberg. Note: Bloomberg estimates for Amkor, TSHT and TFME.

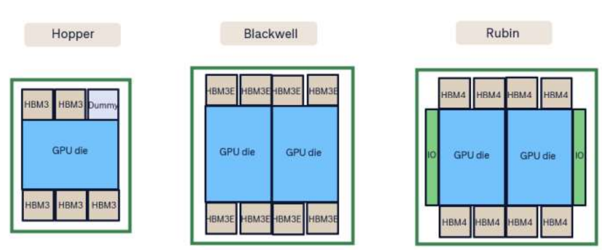
Lengthening test time per chip acts as a multiplier on the equipment build, converting a given level of chip-volume growth into a disproportionately larger test-capacity requirement. A test floor's equipment requirement is a function not only of how many chips must be tested, but of how long each chip occupies a test cell. Here the trend is quite clear: each new generation of AI/HPC chips typically roughly doubles the required test time per device, driven by rising HBM stack counts, higher logic transistor density, more die-to-die interfaces, and a greater number of test insertions. Larger packages compound the effect by reducing parallel test efficiency — Hopper GPU accommodated multiple units per JEDEC standard tray, while Rubin, with its reticle reaching 4-5x, allows only 1-2 units per tray. The combined consequence is that even modest growth in chip output translates into a disproportionately large increase in required test-cell capacity. To absorb this, the OSATs need to build handler and tester capacity faster than chip-unit volume grows.

Figure 30. FT testing time enhancement by each Nvidia AI GPU generation



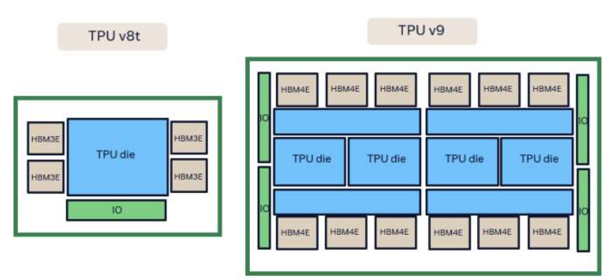
Source: Citi Research estimates

Figure 31. Nvidia's AI GPU packaging architecture



Source: Citi Research estimates

Figure 32. TPU packaging architecture



Source: Citi Research estimates

Global ATE vendors' outlooks independently corroborate the scale of the OSAT equipment build. The capacity guidance of major ATE vendors provides a robust cross-check on the potential OSAT equipment buildout, in our view. Advantest has guided for FY26 sales of ¥1,420bn (+25.8% YoY) and is expanding production capacity from 5,000 in 2026 towards potentially 10,000 test systems in 2028, revising up the SoC tester market to reach around US$9bn in 2026 (+30% YoY vs. 2025), driven by increasing volume and complexity in AI applications. We view these upbeat outlooks as confirmation of a structural OSAT equipment buildout that should continue through at least 2028E, with ATE and handler vendors positioned as direct beneficiaries.

In response, Hon has targeted an aggressive 50% capacity CAGR in the long term. It has acquired a new facility in Daya with potential to expand current capacity by 20-30%. Nonetheless, its current expansion plan still cannot fulfill customer order demand, thus we would not be surprised if Hon announces further new land/facility acquisition to accelerate its expansion. Management has disclosed an internal target of long-term capacity addition by c50%, which we read as strong commitment to long-term project visibility.

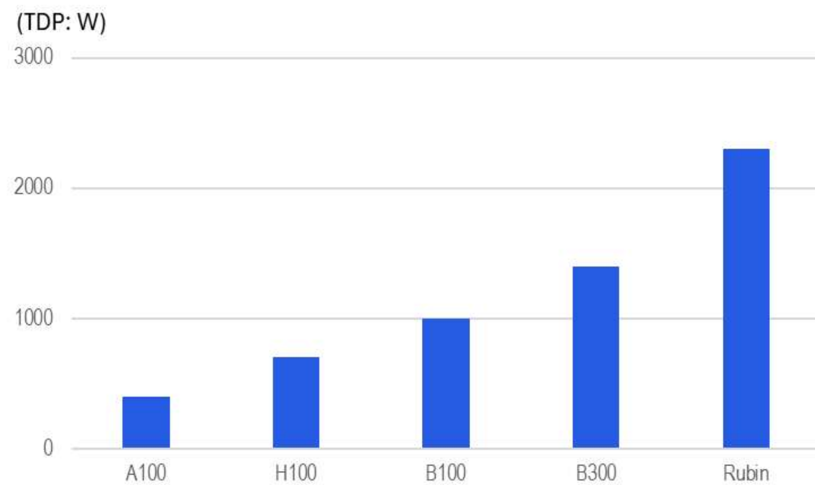
Secular Industry Trend 2

Thermal Control Capability Transitioning

Now a core enabler of smooth testing workflow

Active thermal control (ATC) has shifted from a peripheral handler accessory to the primary performance-enabling component in the AI/HPC final-test flow. The reason is structural: a test result is only valid if the device under test (DUT) is held at its specified operating temperature throughout the test pattern, and as AI chip thermal design power (TDP) escalates generationally, conventional air-cooled handlers can no longer remove heat fast enough to maintain that stability. A decade ago, a typical packaged device drew only single-digit watts; today's AI accelerators routinely sit in the kilowatt range, and the roadmap points towards devices several times beyond that (>5kW). The migration towards 3D-stacked architectures — including NVIDIA's Feynman platform — further intensifies the challenge, as heat must be removed through the upper dies of a vertical stack. The result is a reordering of customer priorities: mechanical handler performance has become a baseline requirement, while thermal management capability now appears the decisive differentiator in equipment selection.

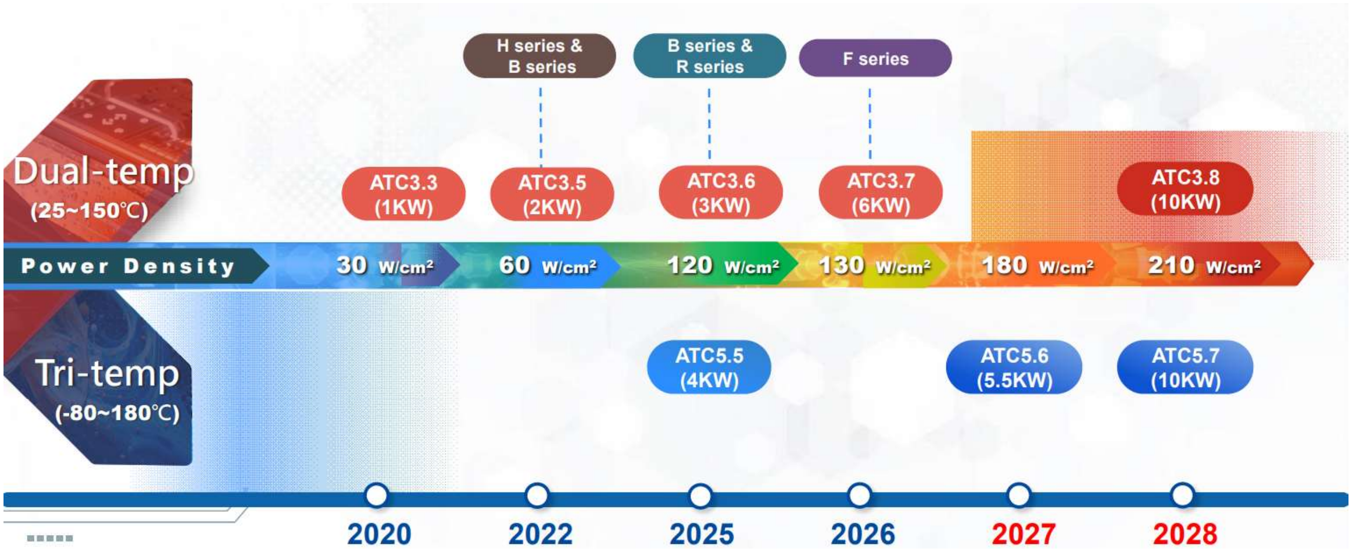
Figure 33. Increasing TDP by each Nvidia AI GPU generation



Source: Citi Research estimates

Hon Precision's ATC roadmap is explicitly synchronized with customers' chip roadmaps, climbing from 1kW in 2020 towards 10kW by 2028 — Hon Precision operates what we view as the industry's broadest ATC portfolio, spanning both dual-temperature and tri-temperature systems. Per company data, the dual-temp ATC roadmap is expected to advance from 3kW (ATC3.6) in 2025 to 6kW (ATC3.7) in 2026-27, with 10kW (ATC3.8) targeted for 2028. The tri-temp ATC roadmap should advance from 4kW (ATC5.5) in 2025 to 5.5kW (ATC5.6) in 2027 and 10kW (ATC5.7) by 2028, supporting an industry-leading coverage range spanning roughly -80°C to +180°C. The strategic point is that this structural roadmap is not speculative — it is explicitly aligned with customers' chip roadmaps, which, in our view, helps cement Hon Precision's dominant position in AI chip handlers market.

Figure 34. Hon Precision – ATC technology roadmap



Source: Citi Research, Hon Precision

ATC migration could drive >10% ASP upgrade every generation — Typically, Hon Precision bundles its handler with its ATC solutions for clients. By type, its offerings run from around NT$6mn for a handler-only configuration, to NT$12-15mn for a handler bundled with dual-temp ATC, to NT$15-18mn for a handler with tri-temp ATC, with next-generation 10kW tri-temp systems likely exceeding NT$20mn, per our estimates. We model the shipment mix across configurations to remain broadly stable in 2026-27, with dual-temperature ATC-equipped handlers representing ~75% of total shipments, tri-temperature systems at c20%, and handler-only configurations declining to <5%. Note that more than 40% of tri-temperature ATC demand for Hon Precision is now correlated with AI chips, with the remaining ~60% sourced from automotive and LEO satellite applications, per our estimates. With the shift towards higher TDP requirement in each AI accelerators generation, we estimate ASP expansion from ATC migration alone could contribute +10-15% top-line growth as generation migrates.

Cold plate is the second pillar of the thermal value chain, migrating from conventional cold plate to MCCP with continued ASP expansion across the AI chips generation cycle

Cold plates currently contribute low-20% of Hon's total revenue. They serve as a recurring, margin-accretive consumable stream that scales with the installed base. Cold plate replacement demand is driven by customer product generation, since each chip requires a differently shaped cold plate. Across a single handler's service life, a recurring stream of cold-plate purchases can be well expected as a typical 1-2 year usage cycle should drive regular replacement demand. Moreover, with higher temperature control requirement, as chips design evolves to multi-die, we should be seeing more customized cold plate designs and faster replacement demand amid the ramping of AI GPU, ASIC and CPU in the upcoming years.

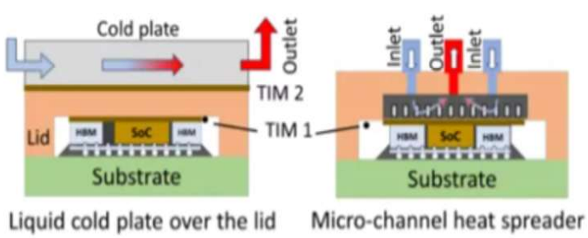
Alongside this replacement cadence, the cold plate is itself moving up a technology curve. Hon Precision's micro-channel design (MCCP), introduced in early 2026, roughly doubles the heat-dissipation density of the conventional ones. Hon disclosed that its MCCP will start shipment from 2Q26 onwards ASP enhancement by 15%, compared to the current cold plates.

We see the cold-plate line as one of the most reliable structural contributors to both revenue growth and margin. The cold-plate revenue trajectory is anchored to two compounding factors: a structurally expanding handler installed base, and the per-unit content uplift as adoption of higher-priced cold plates and MCCP rises. We forecast Hon's cold plates revenue to outpace its overall revenue growth, with contribution rising from low-20% in 2025 to nearly 25% in 2026.

The next frontier is likely the micro-channel lid (MCL)

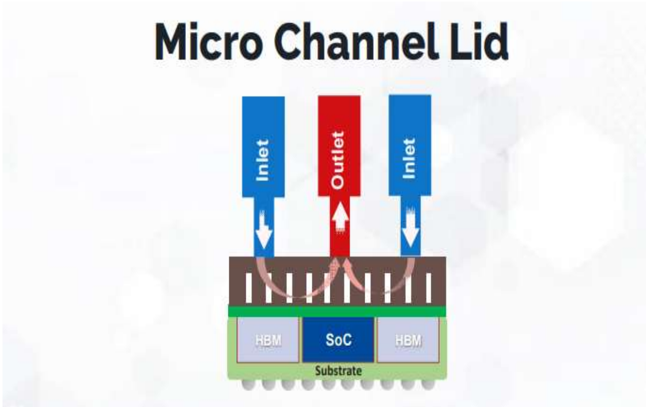
Rather than seating a separate cold plate on the chip, MCL embeds coolant channels within the package lid. Thus, the test handler must automatically connect and disconnect micro-channel water pipes and ensure no leakage occurs during testing – a meaningfully higher bar for handler automation, sealing structure, and thermal control than a conventional cold-plate workflow. In our view, MCL is a 2027-onward optionality driver rather than a 2026 contributor. As per the company, the MCL handler is awaiting MCL-vendor customer qualification. If the final design passes qualification, Hon Precision could deliver a test prototype to the customer within three months. We would treat MCL as upside optionality to the base case, with the key checkpoint being confirmation of customer approval of the lid design and adoption feedback from the supply chains.

Figure 35. Comparison of cold pates/MCCP and MCL



Source: Citi Research, Hon Precision

Figure 36. Hon's MCL product concept



Source: Citi Research, Hon Precision

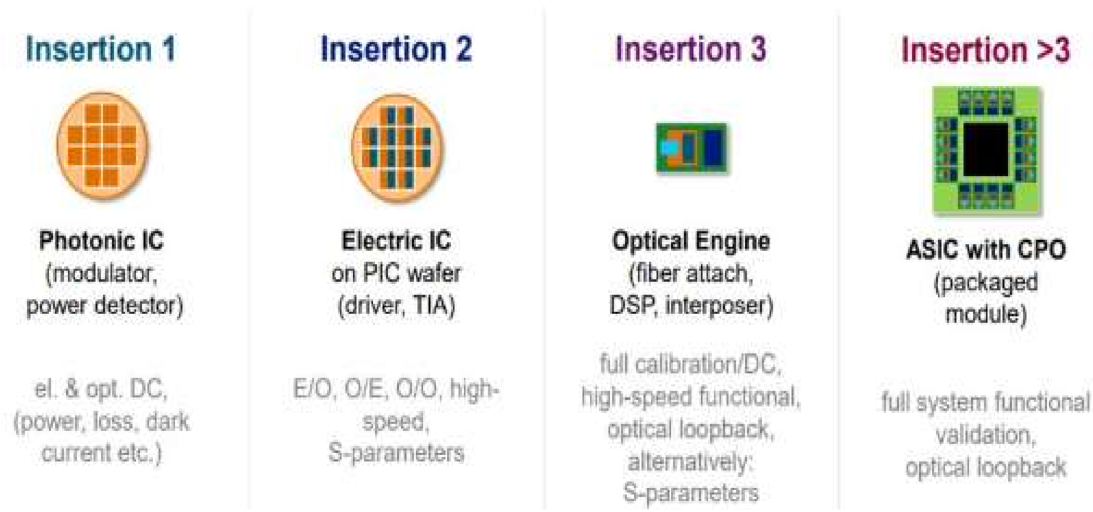
Secular Industry Trend 3

Evolving Chips Packaging Could Create New Platform Opportunity

CPO as one of the biggest growth potential drivers

Co-packaged optics (CPO) integrates photonic integrated circuits and optical engines together with switch ASICs or XPU on a single substrate, replacing the high-power pluggable optical modules used in conventional networking. As CPO fuses optical and electronic components into one package, testing it requires the validation of optical signals — power, insertion loss, modulation efficiency, bit-error-rate, eye-diagram, multi-channel coherence — layered on top of conventional electrical measurement. To manage that complexity, the industry has converged on a multi-stage test-insertion flow (from Insertion 1 to Insertion 4), and Hon Precision's positioning across that flow is, in our view, the single most under-modelled component of the company's medium-term revenue trajectory. We believe CPO, particular for Insertion 4 and FT testing, could drive a structural growth opportunity from 2027 and beyond.

Figure 37. CPO testing workflow



Source: Citi Research, Advantest

Company's role is concentrated at package level (Insertion 4 and CPO FT)

The CPO flow runs from wafer-level photonic and electronic IC test, through optical-engine die-level qualification, to full package-level validation. The first three insertions occur at the wafer or at the die level, where the equipment opportunity flows to probers and specialised optical testers. Hon Precision's role is concentrated at the package level — the current CPO final-test (FT) stage is already in mass production, and the next-stage Insertion 4 optical-electrical co-test is in co-development.

Insertion 4 is the package-level step, performed once the optical engine and the host ASIC have been assembled together onto the substrate. In terms of structure, it looks like a conventional ATE-based final test, but it carries an extra optical dimension that a standard electrical FT does not. The validation is genuinely system-level: the test traces a complete signal loop — confirming that the host ASIC can drive a signal out through the substrate, into the optical engines, and onto the fibres, and back. It also exercises each transmit and receive channel, checks consistency across multiple channels, and holds the package under thermal load — and along the way verifies three things that only surface once everything runs together: whether power delivery holds steady, whether the package stays within thermal limits at full load, and whether the optical engines and the ASIC interfere with one another. This is the insertion that Hon Precision participates in.

Current project: CPO FT, already in mass production, shipments booked

Hon Precision has engaged with a key customer across three CPO switch projects — it has already shipped FT handlers for the electrical test of standalone ASICs, standalone optical engines, and integrated ASIC-plus-OE modules. Hon Precision has shipped more than 100 units of CPO FT electrical handlers to a client and the current order visibility has extended into 1Q27 with products delivery continuing. This CPO handler's ASP is higher than Hon Precision's corporate-average ASP — suggesting the CPO mix shift should lift Hon's blended ASP as it grows.

Future development: Insertion 4, a strategic next step

Validation targeted for 4Q26

The most strategical important development is the CPO Insertion 4 optical-electrical co-test handler. **Insertion 4 co-test combines two already-demanding test loads into a single, longer insertion.** The host ASIC in a CPO package is itself a leading-edge accelerator or switch silicon, and its electrical test time is already long. On top of the full electrical validation of the ASIC, the co-test must also exercise the optical engines — and critically, it does so with the optical engines mounted in place on the package and tested concurrently with the ASIC.

Longer test time per device means each Insertion 4 unit ties up a handler for longer, which — for a given volume of CPO packages — raises the number of handlers required. Combined with new hardware that Insertion 4 demands (the optical alignment stage, external light source test head, and dedicated load board that a standard FT handler lacks), this is what makes the Insertion 4 handler both higher-ASP and higher-unit-intensity than a conventional electrical FT handler. We expect the Insertion 4 optical-electrical handler could carry an ASP higher than Hon's current premium product lines, to be over NT$20mn, given its totally new equipment design as it requires an external light source test head and an optical alignment stage and more active thermal management. In our view, the lengthening of test time at Insertion 4 and the higher ASP profile should serve as a key growth opportunity for Hon Precision.

Hon Precision has been the leading FT handler developer for Insertion 4 optical-electrical co-test of CPO modules, working in close collaboration with an ATE vendor and the end customer. We are confident in Hon Precision's potential Insertion 4 leadership, as optical alignment requires sub-micron mechanical precision with vibration isolation and active thermal compensation, while few competitors possess the cross-disciplinary engineering muscle to replicate this in a high-volume manufacturing environment. Moreover, Insertion 4 qualification is likely to be lengthy given the safety-critical nature of CPO products in data-centre networks; customers who have invested the time qualifying Hon's platform are less likely to repeat the exercise with an alternative vendor, in our view.

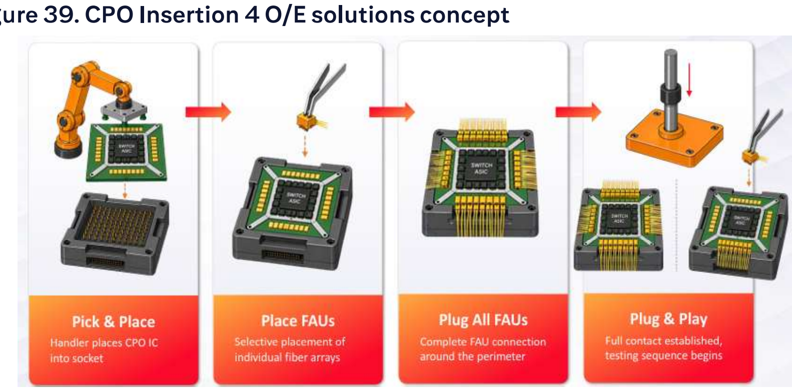
While the final configuration is still not yet finalized, the company targets verification of the optical-electrical handler to enter the final stage in 4Q26, with mass production/initial deliveries in 2027.

Figure 38. Hon – CPO testing solutions (Insertion 4 and CPO FT)

Test Insertion	Insertion 4	CPO FT		
	Joint Development	Mass Production		
Test Content	ASIC +Optical Engine	ASIC +Optical Engine	ASIC	Optical Engine
	O/E	E/E	E/E	E/E
HON.PREC Test Equipment	ATE Tester + ATC Handler			
	3000W	2000W	1000W	1000W
	Single Site	Single Site	Single/Dual/Quad Site	Dual Site
	Optical Alignment Stage; ASIC+OE Load Board	ASIC+OE Load Board	ASIC Load Board; ASIC ASIC Load Board; ASIC ASIC ASIC ASIC Load Board	OE OE Load Board

Source: Citi Research, Hon Precision

Figure 39. CPO Insertion 4 O/E solutions concept



Source: Citi Research, WinWay

Beyond CPO, larger packaging driving larger tray spec requirement

We also expect Hon Precision to benefit from the rising IC package size that requires entirely new handler platforms. The legacy JEDEC standard tray (135.9×322.6mm^2) holds only two to three packaged ICs for the Hopper and Blackwell platforms. The Rubin generation should still fit within the JEDEC matrix standard, but the Feynman generation is likely to outgrow it. As next-generation AI GPUs/ASIC could go beyond the current tray platform package size limit of 120x150mm^2, an entirely new, larger tray system is needed. Hon Precision recently announced a large-package handler platform. Its tray size is as large as 380x380mm to accommodate ICs' package size to 250x250mm^2 with contact force being able to reach 2x increase to 2,000kg because a larger package needs far more even pressure when it is pressed onto the test socket. With new design of larger tray, higher contact force, ATC system support, and full automation, we estimate new large-package handler ASP could carry a 20-30% premium over the current dual-temp ATC handler.

As AI chips packaging continues to evolve and more larger packaging solutions are required, such as CoPoS, SoIC, etc., we expect OSATs to be driven to adopt Hon's entirely new larger tray handler platform, creating a wave of new equipment pull-in demand for Hon. We expect initial revenue contribution to emerge in 2027 and before turning more meaningfully thereafter. The company plans to complete the new handler design and validation in 1H26 and begin mass production for major HPC customers from 2H26, deliberately pre-building capacity ahead of next-generation AI chips.

Figure 40. Handler tray specs comparison

Try generation	Try spec	Package size	Reticle size
Legacy JEDEC tray	322.6x135.9mm	<120x150mm	<5x
Hon's newly large-packaged tray	380x380mm	up to 250x250mm	up to 10x

Source: Citi Research, Hon Precision

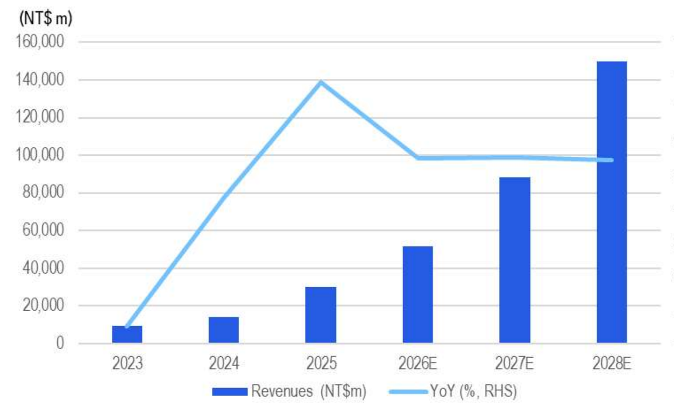
Financials Analysis and Forecast

Strong revenue acceleration and margin expansion

In 2024 and 2025, Hon Precision delivered strong revenue growth and structural margin expansion, driven by a clear transition into the AI/HPC test-equipment pull-in cycle. Revenue growth accelerated from 47% in 2024 and 116% in 2025 — indicating the business more than tripled in two years.

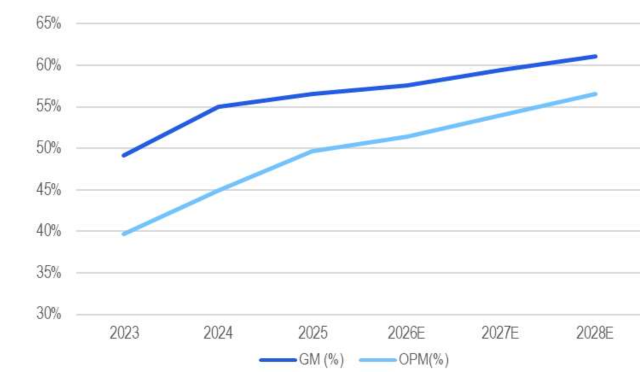
Gross margin rose in parallel, from 49.2% in 2023 to 55.0% in 2024 and 56.5% in 2025. This reflects three structural drivers, rather than a one-off cyclical upturn, in our view.

Figure 41. Hon - revenue (NT$ mn) & YoY growth



Source: Citi Research, Company Reports

Figure 42. Hon - GM and OPM trends



Source: Citi Research, Company Reports

Strong shipment growth on a clear customer-mix transition towards leading AI/HPC. The first and most important driver, in our view, is that Hon Precision's customer mix has shifted decisively into the highest-value layer of the semiconductor end-market. AI/HPC/ASIC applications accounted for 78% of 1Q26 order booking, up from broadly mid-60% in 1H25 and ~73% in 2024. The volume driver has been a rapid back-end ramp of leading AI accelerator testing capacity — including major US GPU vendors, AI ASICs — all of which require Hon Precision's FT handlers paired with ATC systems. As these customers typically purchase higher-specification configurations than the legacy mobile and consumer chip base that dominated Hon Precision's revenue prior to 2023, the mix shift has lifted both unit volumes and the average dollar content per shipment of the company.

Specification migration has been driving structural ASP and content uplift. The second driver is content expansion within each handler shipment. As AI chip TDP has continued to climb, Hon Precision's customers have upgraded along the ATC ladder. The result is that Hon Precision is capturing more dollars per handler with each successive AI chip generation, on top of unit-volume growth — a dynamic that has contributed materially to both revenue growth and gross margin, since ATC content carries structurally higher margins than the handler unit alone.

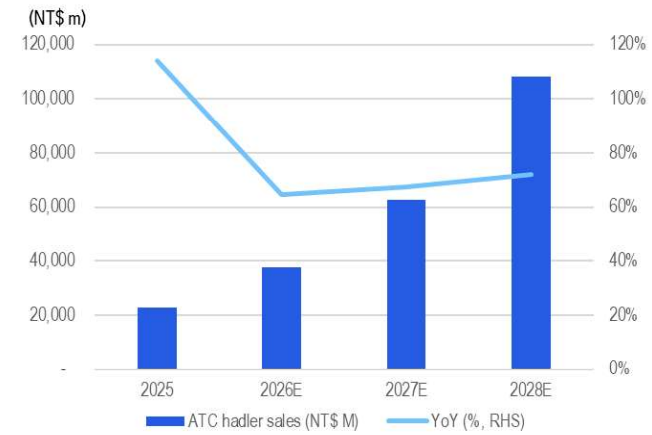
Cold plates as a recurring revenues and margin driver. Hon has been seeing strong growth in its high-margin cold pate product line. Cold plates now contribute a low-

20s percentage of revenue, and because the cold plate is a chip-specific consumable that is replaced every 1-2 years against a handler's near-decade service life, it provides a recurring revenue stream that scales with the installed base, rather than only with new equipment shipments. Together with more customized requirements by high-end chip designs, more premium cold plate solutions are likely needed, which should support Hon's revenue trajectory and margin profile.

Outlook

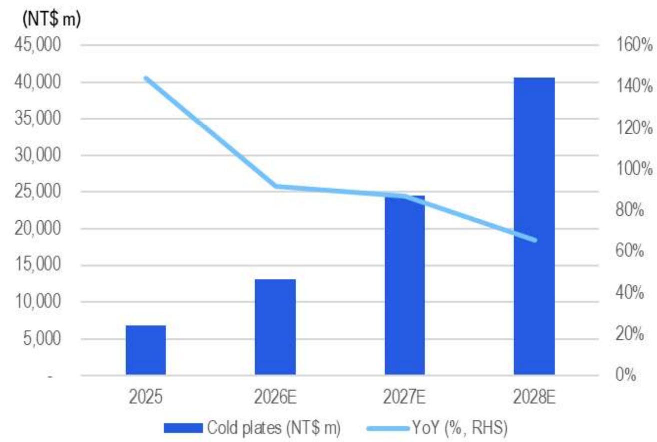
Looking ahead — we see Hon's 2024-25 acceleration extending to 2026-28E, supported by its aggressive capacity expansion, more content growth and new opportunities in evolving packaging. We forecast its sales/net profit to grow at CAGRs of c70/77% over 2025-28, as we expect expanding AI chip projects (AI GPUs, AAIC, CPUs, etc.) and lengthening test time to boost multi-year handler buildout demand. Moreover, larger packages, newly integrated functions (AOI) and continued upgrade demand for ATC systems should drive structural ASP expansion of roughly 10-15% per annum, on our estimates. Lastly, evolving packaging architectures could open entirely new opportunities. We believe the CPO evolution and development in Hon's Insertion 4 optical-electrical co-test handler could drive significant upside potential to its long-term outlook, and positive development in the MCL technology could provide another upside catalyst. To meet this demand, Hon Precision has raised its 2026 capacity addition target to more than 40% YoY, with an internal project targeting a further ~50% expansion in the long term, and the Longshan facility is scheduled for 2028.

Figure 43. Hon – ATC handler revenue (NT$ mn) & YoY growth



Source: Citi Research, Company Reports

Figure 44. Hon – cold plates revenue (NT$ mn) & YoY growth



Source: Citi Research, Company Reports

Financial forecasts summary

Figure 45. Hon – financial forecast summary

	2025				2026				2027							
(NT$ in Mn, year-end Dec)	1Q	2Q	3Q	4Q	1Q	2QE	3QE	4QE	1QE	2QE	3QE	4QE	2025	2026E	2027E	2028E
Revenue	**5,913**	**6,885**	**8,198**	**9,276**	**10,725**	**11,627**	**13,514**	**15,749**	**17,671**	**20,635**	**23,458**	**26,585**	**30,271**	**51,615**	**88,350**	**149,816**
COGS	-2,419	-2,847	-3,468	-4,423	-4,693	-4,939	-5,720	-6,514	-7,227	-8,423	-9,607	-10,616	-13,157	-21,866	-35,874	-58,308
Gross Profit	**3,494**	**4,037**	**4,730**	**4,852**	**6,032**	**6,689**	**7,794**	**9,234**	**10,444**	**12,212**	**13,851**	**15,969**	**17,114**	**29,749**	**52,476**	**91,509**
Operating Expense	-360	-394	-565	-749	-672	-541	-790	-1,201	-976	-760	-1,148	-1,869	-2,068	-3,204	-4,753	-6,853
SG&A expenses	-218	-185	-276	-398	-315	-255	-387	-598	-430	-349	-554	-876	-1,077	-1,554	-2,209	-2,994
R&D expenses	-208	-163	-239	-430	-374	-241	-354	-682	-563	-365	-543	-1,072	-1,039	-1,651	-2,544	-3,859
EBIT	**3,133**	**3,643**	**4,166**	**4,104**	**5,360**	**6,148**	**7,003**	**8,034**	**9,467**	**11,452**	**12,704**	**14,101**	**15,046**	**26,545**	**47,723**	**84,656**
Pre-Tax Profit	3,235	3,156	4,524	4,669	5,798	6,395	7,209	8,309	9,902	11,729	12,939	14,406	15,584	27,711	48,976	85,869
Tax	-667	-690	-907	-958	-1,174	-1,398	-1,446	-1,704	-2,005	-2,565	-2,595	-2,955	-3,222	-5,723	-10,120	-17,742
Net Profit	**2,568**	**2,466**	**3,617**	**3,711**	**4,624**	**4,996**	**5,763**	**6,605**	**7,898**	**9,164**	**10,344**	**11,451**	**12,362**	**21,988**	**38,856**	**68,127**
EPS (NT$)	**15.89**	**15.26**	**22.39**	**22.73**	**25.70**	**27.77**	**32.03**	**36.71**	**43.90**	**50.94**	**57.50**	**63.65**	**76.30**	**122.22**	**215.98**	**378.69**
Margins (%)																
Gross Margin	59.1%	58.6%	57.7%	52.3%	56.2%	57.5%	57.7%	58.6%	59.1%	59.2%	59.0%	60.1%	56.5%	57.6%	59.4%	61.1%
Operating Margin	53.0%	52.9%	50.8%	44.2%	50.0%	52.9%	51.8%	51.0%	53.6%	55.5%	54.2%	53.0%	49.7%	51.4%	54.0%	56.5%
Net Margin	43.4%	35.8%	44.1%	40.0%	43.1%	43.0%	42.6%	41.9%	44.7%	44.4%	44.1%	43.1%	40.8%	42.6%	44.0%	45.5%
Sequential Growth (%)																
Revenue	19%	16%	19%	13%	16%	8%	16%	17%	12%	17%	14%	13%	116%	71%	71%	70%
Gross Profit	30%	16%	17%	3%	24%	11%	17%	18%	13%	17%	13%	15%	122%	74%	76%	74%
EBIT	43%	16%	14%	-1%	31%	15%	14%	15%	18%	21%	11%	11%	139%	76%	80%	77%
Net Profit	31%	-4%	47%	3%	25%	8%	15%	15%	20%	16%	13%	11%	134%	78%	77%	75%
EPS	30%	-4%	47%	2%	13%	8%	15%	15%	20%	16%	13%	11%	134%	60%	77%	75%

Source: Citi Research, Hon Precision

Figure 46. Hon – Citi estimates vs. consensus

	2026E			2027E			2028E		
(NT$mn)	Citi	Con.	Diff.	Citi	Con.	Diff.	Citi	Con.	Diff.
Sales	51,615	51,831	0%	88,350	85,434	3%	149,816	142,558	5%
Gross profit	29,749	29,668	0%	52,476	49,801	5%	91,509	85,672	7%
Operating profit	26,545	26,290	1%	47,723	45,862	4%	84,656	73,768	15%
Net income	21,988	21,702	1%	38,856	37,405	4%	68,127	63,262	8%
EPS (NT$)	122.22	124.66	-2%	215.98	204.70	6%	378.69	347.38	9%
Gross margin	57.6%	57.2%	+0.4 ppt	59.4%	58.3%	+1.1 ppt	61.1%	60.1%	+1.0 ppt
Operating margin	51.4%	50.7%	+0.7 ppt	54.0%	53.7%	+0.3 ppt	56.5%	51.7%	+4.8 ppt
Net margin	42.6%	41.9%	+0.7 ppt	44.0%	43.8%	+0.2 ppt	45.5%	44.4%	+1.1 ppt

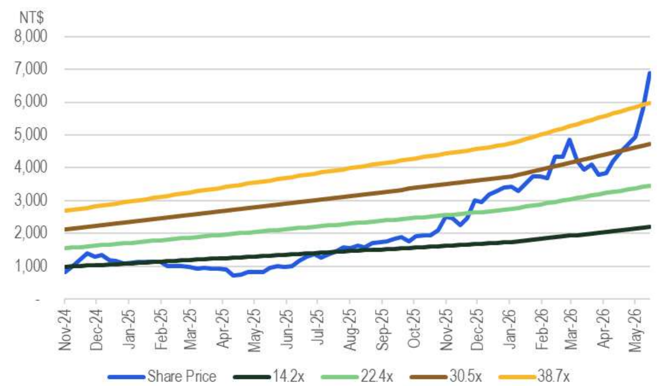
Source: Citi Research, Bloomberg estimates

Valuation Analysis

NT$11, 000 target price

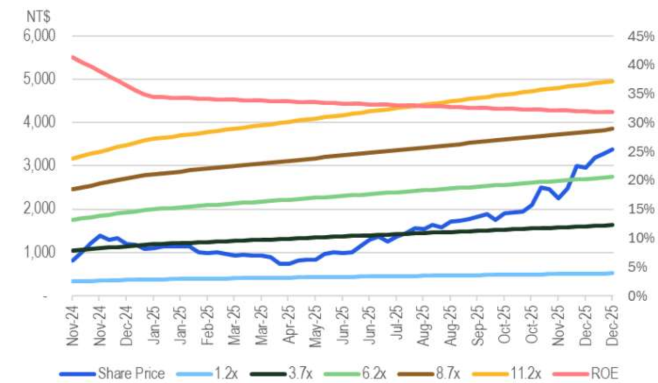
Our target price for Hon of NT$11,000 is based on a target PER multiple of 36x applied to the average of our 2027E/28E EPS estimates. Our PER target of 36x is set at 1-std above its 1-year PER average, justified by its high share of the AI/HPC testing handler market, which we estimate could support an earnings CAGR of 77% in 2025-28, higher than its 3-year CAGR at c40% in 2022-25. Our estimated 77% earning CAGR is also higher than the +40% average (Bloomberg consensus) of its global testing equipment peers. As its peers have been trading at average 36x PER on 2027/28 consensus EPS in the past 1 year, we believe Hon Precision deserves a further re-rating to our target PER of 36x. We also note QFII holding on Hon Precision is currently lower than the peer average of 34%. At our target price, the shares would trade at 2027E/28E P/B of 17x/11x.

Figure 47. Hon – PE band



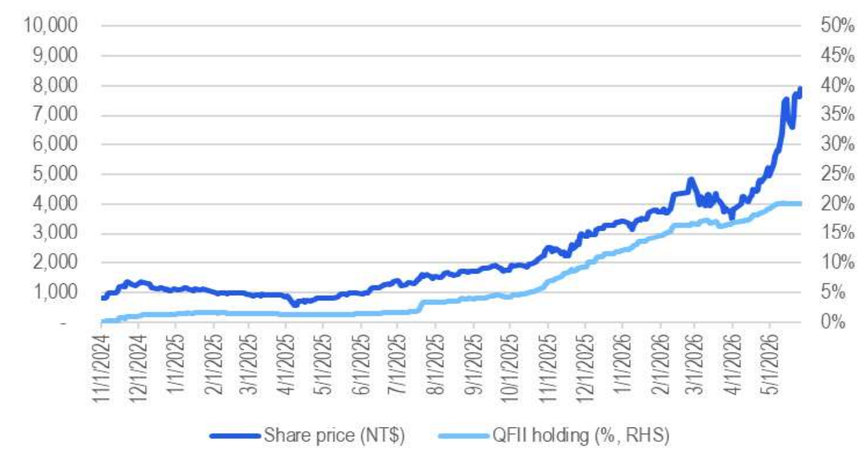
Source: Citi Research, Bloomberg

Figure 48. Hon – PB band & ROE trend



Source: Citi Research, Bloomberg

Figure 49. Hon – further share price re-rating potentials as QFII holding is still low at 20% now



Source: Citi Research, TEJ

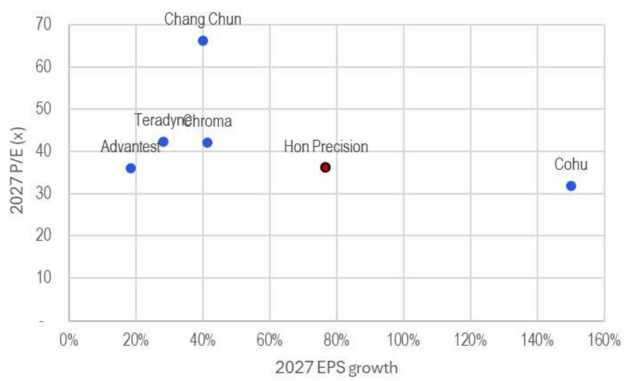
Peer valuations

Figure 50. Hon – peers valuation comparison

Ticker	Company	Rating	Price 5/27/2026		Mkt Cap (US$m)	EPS (local Currency) 2027	EPS (local Currency) 2028	P/E (x) 2027	P/E (x) 2028	P/B (x) 2027	P/B (x) 2028	ROE (%) 2027	ROE (%) 2028	Dividend Yield(%) 2027	Dividend Yield(%) 2028	EPS Growth (%) 2027	EPS Growth (%) 2028
7769.TW	**Hon Precision**	**1**	**TWD**	**7,800**	**44,605**	**216.0**	**378.7**	**35.7**	**20.3**	**11.8**	**34.5**	**39.0**	**31.5**	**1.7**	**0.7**	**76.7**	**75.3**
2360.TW	Chroma	1	TWD	2,625	35,478	62.2	75.2	41.3	34.2	28.0	30.0	71.4	75.3	1.2	0.8	41.2	20.9
6857.JP	Advantest	1	JPY	27,130	123,550	753.9	817.1	34.9	32.2	17.6	53.2	53.9	47.9	0.4	n.m.	18.4	8.4
TER.US	Teradyne	1	USD	376	58,833	9.2	10.9	40.9	34.3	13.0	19.8	31.0	27.3	0.2	0.1	28.0	19.1
COHU US	Cohu	NR	USD	50	2,195	1.5	2.2	31.9	21.9	2.7	2.4	27.2	-	n.a.	n.a.	150.0	19.0
300604CH	Changchuan	NR	CNY	220	22,035	3.6	-	65.9	n.m.	17.6	n.a.	28.7	-	0.2	n.a.	40.0	n.m.
Global testing equipment peer average								**43.0**	**30.7**	**15.8**	**26.3**	**42.5**	**50.2**	**0.5**	**0.4**	**55.5**	**16.8**
6223.TW	MPI	1H	TWD	6,390	19,899	104.0	177.5	56.9	33.3	20.5	13.1	39.8	44.2	1.0	1.5	74.9	70.6
6510.TW	CHPT	1H	TWD	3,195	3,330	122.2	148.6	26.9	22.1	7.1	11.9	31.4	28.3	1.6	0.5	99.9	21.7
6515.TW	WinWay	1H	TWD	9,795	11,220	216.5	365.3	42.1	25.0	20.9	27.9	58.4	55.2	1.2	0.5	116.4	68.7
Taiwan testing interface peer average								**41.9**	**26.8**	**16.2**	**17.6**	**43.2**	**42.6**	**1.3**	**0.9**	**97.1**	**53.7**

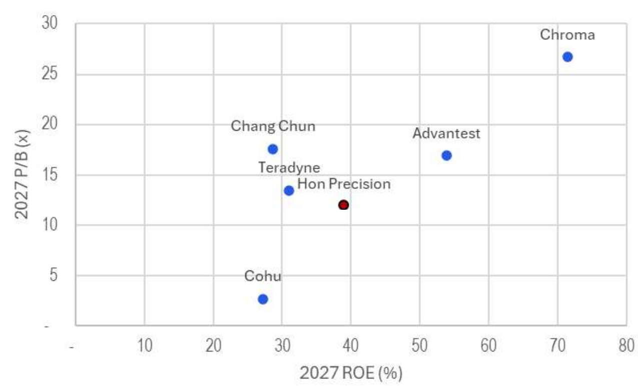
Source: Citi Research, Bloomberg. Note: Consensus estimates for non-rated stocks.

Figure 51. Hon – peer EPS growth & PER comparison



Source: Citi Research, Bloomberg

Figure 52. Hon – peer ROE & PBR comparison



Source: Citi Research, Bloomberg

Key Risks

Our quantitative risk rating system, which looks at historical share price volatility, rates Hon Precision High Risk. However, we are not assigning a High Risk rating as we forecast an earnings CAGR of >75%, significantly stronger than the growth outlook for its global peers. Also, we note its dominant position in AI/HPC handler market and expanding market TAM opportunity, which support our constructive view over its growth outlook.

Key downside risks that could impede the shares from reaching our target price include:

- **Global AI demand slowdown:** Any overall AI market demand weakness would result in slower semiconductor equipment capacity build at Hon's clients, and in turn lead to slower orders for Hon.
- **Unexpected share losses at its major AI projects:** Unexpected issues, including slower product delivery and clients' dual/multi-sourcing strategy, may result in some share loss. We view this risk as low, as it typically requires >1 year for qualification of new equipment vendors to enter the supply chain. The rapid AI chip product generation cycle makes vendor switching even less likely.
- **Slower-than-expected capacity ramp at its new facilities:** Slower-than-expected capacity ramp may result in share loss, as Hon's current capacity is still not enough to fulfill the strong demand. It may also take the company more time than expected to find or secure new sites/plants.
- **Slower-than-expected development of its CPO business:** Any delay in CPO Insertion 4 development may lower market expectations on Hon Precision's long-term growth potential in this area.
- **Increasing competition from new entrants.** Emergence of any new players, including ATE vendors or others, might result in unexpected pricing competition, pressuring Hon's margin outlook.
- **Shorter-than-expected testing time:** Unexpected reduction in testing time, be it due to hyperscalers' changing requirement/industry practice or setbacks in chip advancement, would pose risk of slowing the equipment capacity build by OSAT vendors.
- **Lower-than-expected adoption of high-premium cold plate products:** Slower-than-expected adoption of MCCP would lead to weaker content growth and margins expansion outlook for Hon.

Company background

Founded in 1999, Hon Precision's predecessor, Hon Tech, began operating in Taichung, Taiwan, as a specialist in automated semiconductor test handlers. For years, Hon Tech had focused on developing solutions for the OSAT industry. The company took orders from local Taiwanese IC designers and OSATs working on low-end consumer chips at the beginning. Margins were thin, but the work served as a training ground for automation and handling fundamentals that would later differentiate the company. The company's breakthrough into global accounts came when Hon Precision's reference base widened and its 24-hour customer-service model — engineers are stationed at customer sites, debugging production lines in real time — gave it credibility in the industry.

Nowadays, Hon Precision is a widely recognized supplier of semiconductor IC testing equipment. The company delivers comprehensive, one-stop integrated solutions encompassing IC test handling, ATC, and diverse semiconductor-related equipment. With a wide global footprint, Hon Precision's clientele spans end-users and OSAT providers across Europe, the US, Israel, mainland China, Japan, South Korea, and Southeast Asia.

The company was listed on Taiwan's Emerging Stock Board in October 2024 and moved to the TWSE main board in November 2025.

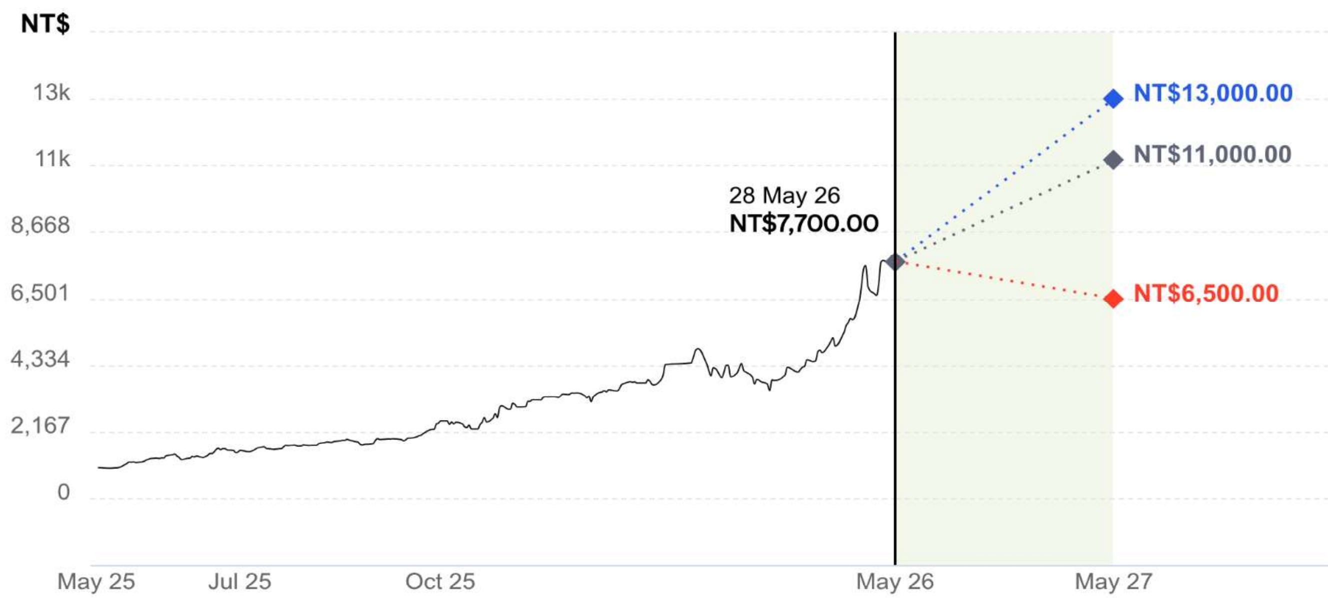
Adding Upside 90-Day Catalyst Watch on Hon Precision (7769.TW)

Direction:	**Upside**
Duration:	Within 90 Days
Catalyst:	Thematic driven

We are constructive on Hon Precision's earnings growth outlook driven by its solid engagement in global leading AI projects. For 2Q earnings, our estimates are higher than consensus as we expect higher margins from a better sales mix and operating leverage, which we believe could drive strong >100% YoY earnings growth. Also, though a long-term opportunity, we believe co-packaged optics (CPO) engagement with key customers could also drive significant upside potential.

Bull/Bear: Hon Precision (7769.TW)

NT$13,000.00 ▲69% Upside	NT$11,000.00 ▲43% Upside	NT$6,500.00 ▼16% Downside



Spread 84pp
Current Price and expected returns (upside/downside) as of 28 May 2026

BULL Assumptions

- New growth opportunity in new projects
- Faster-than-expected capacity ramp for its new facilities
- Faster-than-expected development in its CPO business

BASE Assumptions

- Solid 3-year sales CAGR of ~70%
- Strong 3-year earnings CAGR of >75%

BEAR Assumptions

- Unexpected share loss at its major AI projects
- Slower-than-expected capacity ramp for its new facilities
- Weaker-than-expected development in its CPO business

Hon Precision

Company description

Hon Precision is a leading Taiwanese manufacturer specializing in IC semiconductor automation equipment. Headquartered in Taichung City, Taiwan, it is most known globally for providing high-precision IC test handlers, temperature control solutions, and complete automated integrated solutions for major OSAT and IDM customers.

Investment strategy

We have a Buy rating on Hon Precision as we view it sitting in a structurally protected position at the heart of the AI/HPC back-end test cycle, with growth visibility extending well into the next few years — the company holds >90% share of the AI/HPC FT handler market and 60-70% of the ASIC SLT market. We forecast a solid strong sales revenue CAGR in 2025-28E with gross margin continuing to expand. In our view, the binding constraint on Hon Precision's near-term trajectory is now capacity, rather than demand — placing the company in the early phase of a multi-year structural growth and margin-expansion cycle.

Valuation

Our target price of NT$11,000 is based on a target PER multiple of 36x applied to the average of our 2027E/28E EPS estimates. Our PER target of 36x is set at 1-std above its 1-year PER average, justified by its high share of the AI/HPC testing handler market, which we estimate could support an earnings CAGR of 77% in 2025-28, higher than its 3-year CAGR at c40% in 2022-25. Our estimated 77% earning CAGR is also higher than the +40% average (Bloomberg consensus) of its global testing equipment peers. As its peers have been trading at average 36x PER on 2027/28 consensus EPS in the past 1 year, we believe Hon Precision deserves a further re-rating to our target PER of 36x.

Risks

Our quantitative risk rating system, which looks at historical share price volatility, rates Hon Precision High Risk. However, we are not assigning a High Risk rating as we forecast an earnings CAGR of >75%, significantly stronger than the growth outlook for its global peers. Also, we note its dominant position in AI/HPC handler market and expanding market TAM opportunity, which support our constructive view over its growth outlook.

Key downside risks that could impede the shares from reaching our target price: 1) Global AI demand slowdown driving weaker capacity build for semiconductor equipment; 2) unexpected share losses at its major AI projects; 3) slower-than-expected capacity ramp at its new facilities, or the company taking a long time to find/secure new sites; 4) slower-than-expected development of its CPO business; 5) shorter-than-expected testing time leading to lower demand for its products; 6) weak adoption of high-premium cold plate products; and 7) increasing competition from new entrants.

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Appendix A-1

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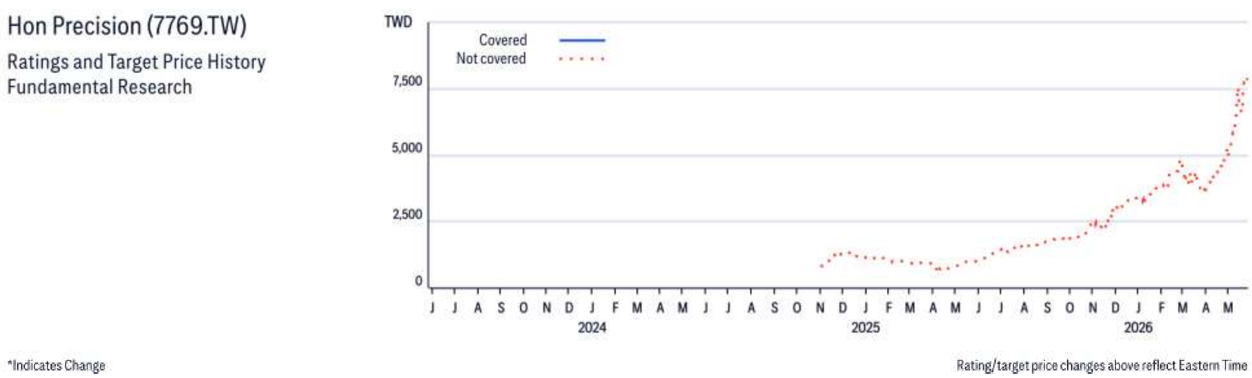

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